A Multifaceted Analysis of Sri Lankan Payment Adoption, Borrowing Behavior, and Financial Clustering

STAT 32663 - Corporate Capstone Project

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By

Group Number 04

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Declaration

We do hereby declare that the work reported in this report which titled as A Multifaceted Analysis of Sri Lankan Payment Adoption, Borrowing Behavior, and Financial Clustering" is originally prepared by us, Group 04 in the purpose of partial fulfilment of requirement of the Bachelor of Science Degree, Department of Statistics & Computer Science, Faculty of Science, University of Kelaniya, Sri Lanka and not for any other academic purposes. No part of this report has been submitted earlier or concurrently for the same or any other degree.

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1. Introduction

1.1 Background

Financial inclusion means people can access and use formal financial services like bank accounts, digital payments, and loans. It is crucial for both promoting economic growth and lowering inequalities.

Sri Lanka has made good progress in financial inclusion. Most adults have bank accounts, and mobile phone ownership is high. However, using digital financial services regularly is still low.

The Global Findex database provides data on how people use financial services. This study uses the 2024/2025 Global Findex microdata for Sri Lanka, with 1,000 respondents.

1.2 Problem Statement

Most people don't have clear ideas about digital banking and financial activities. So, Sri Lanka still has a few issues with financial inclusion:

Problem	Description
Low digital adoption	Only about 48% use digital payments, and even fewer use them daily.
Unequal credit access	Rural and low-income groups depend on informal loans (family, friends, moneylenders)
Financial exclusion	Many people lack bank accounts or saving opportunities, especially the elderly
No targeted services	Financial products are not designed for different customer needs
Access vs. usage gap	High account ownership (82.6%) does not mean active use of digital services

Table 1:Research Problem Description

These problems limit people's ability to save safely, access credit, and participate in the modern economy.

1.3 Research Objectives (Four Pillars)

Pillar	Objective
Pillar 1	Analyze digital payment adoption patterns and what factors influence adoption
Pillar 2	Examine borrowing behavior and credit access across different groups
Pillar 3	Investigate financial inclusion patterns and saving behavior
Pillar 4	Identify distinct customer segments in Sri Lanka using clustering techniques

1.4 Research Questions

This study answers the following questions:

For Pillar 1 (Digital Payments):

I. What percentage of Sri Lankans use digital payments?
II. Which demographic groups (age, gender, education, income) are more likely to adopt digital payments?
III. What are the main barriers to digital payment adoption?

For Pillar 2 (Borrowing Behavior):

I. Who borrows money in Sri Lanka?
II. Why do people borrow (purposes)?
III. Do people use formal or informal sources for borrowing?

For Pillar 3 (Financial Inclusion):

I. How many people have bank accounts across different groups?
II. What are the saving patterns in Sri Lanka?
III. How does mobile money usage look like?

For Pillar 4 (Customer Segmentation):

I. What distinct financial behavior segments exist in Sri Lanka?
II. How can financial services be tailored for each segment?

1.5 Significance of Study

This study provides valuable insights for multiple stakeholders in Sri Lanka's financial sector. For policymakers at the Central Bank of Sri Lanka and government agencies, the findings offer

evidence-based guidance for designing financial inclusion strategies and digital infrastructure investments. Financial institutions can use the customer segmentation results to develop targeted products that meet the specific needs of different population groups, from digital elites to the unbanked.

Most of the companies will benefit from understanding adoption barriers and market opportunities, particularly the large, untapped potential for daily digital payment solutions. For academic researchers, this study contributes Sri Lanka-specific knowledge to the global financial inclusion literature, challenging assumptions about urban-rural divides and confirming the importance of education and income as adoption drivers. This study is helping Sri Lanka build a more inclusive digital economy.

2.Literature Review

2.1 The Access-Usage Gap

A consistent finding across financial inclusion literature is the distinction between having access to financial services and actively using them. Allen et al. (2016) and the World Bank's Global Findex reports document that high account ownership does not guarantee active digital payment usage. Many account holders maintain inactive accounts or continue relying on cash for daily transactions. This "access-usage gap" is attributed to factors including lack of trust, perceived cost, limited digital literacy, and insufficient merchant acceptance. In Sri Lanka, Gunasinghe & Perera (2022) found account ownership at 89% but digital payment usage at only 45%, confirming this pattern. This study aims to quantify Sri Lanka's access-usage gap and identify whether the main barriers are trust, cost, usability, or simply no perceived need for digital transactions.

2.2 Borrowing Behavior Patterns

Research on borrowing behavior shows distinct patterns across demographic groups. The life-cycle hypothesis (Modigliani & Brumberg, 1954) suggests that borrowing peaks in middle age (35-54 years) when household formation, business investment, and children's education create financial needs. Banerjee & Duflo (2007) found that borrowing does not always increase with poverty, the poorest often cannot access formal or informal credit due to perceived risk. Regarding borrowing sources, Karlan et al. (2019) distinguish between formal borrowing (banks, financial institutions) and informal borrowing (family, friends, moneylenders), with informal sources playing a larger role where formal access is limited. This study verifies this "inverted U" borrowing pattern in Sri Lankan data and links loan purposes (business, health, education) to different demographic groups.

2.3 Digital Payment Adoption

Most recent research suggests that urban-rural digital gaps can be relatively small, implying that behavioral factors and habits are bigger barriers than infrastructure. Withana & Rathnayaka (2021) applied the UTAUT model to Sri Lankan young adults and found that performance

expectancy and social influence mattered more than geographic location. This suggests that encouraging small, frequent digital transactions can build lasting habits regardless of where people live. This study compares adoption rates between urban and rural Sri Lankans and examines whether infrastructure or behavior presents the greater barrier to digital payment adoption.

3.Methodology

The methodological approach used to analyze financial inclusion patterns, digital payment adoption, borrowing behavior, and customer segmentation in Sri Lanka. The methodology is designed to address the four research objectives using quantitative analysis of Global Findex microdata.

3.1 Data Source

The Global Findex database is the world's most comprehensive dataset on financial inclusion, collected by the World Bank. It provides:

- Nationally representative surveys of adults (age 15+)
- Standardized indicators across economies
- Detailed information on account ownership, payments, saving, borrowing, and financial resilience

Characteristic	Description
Dataset	findexmicrodata2025_labelled.csv
Country	Sri Lanka
Sample size	1,000 respondents
Year	2024/2025
Initial variables	78 features
Population represented	Sri Lankan adults (age 15+)

Table 2:Sri Lanka Sample

The 78 variables fall into several categories:

Category	Description	Count
Demographics	Age, gender, education, income, employment, location	~10
Account ownership	Bank accounts, mobile accounts, digital accounts	~8
Payment behavior	Digital payments, merchant payments, bill payments	~15
Saving behavior	Saved, saving methods, saving purposes	~8
Borrowing behavior	Borrowed, sources, purposes	~10
Income receipt	Wages, government payments, agricultural payments	~12
Financial confidence	Capability, confidence, resilience	~15

3.2 Data Preprocessing

3.2.1 Data Loading and Validation

Initial steps included:

- Loading the dataset for Sri Lanka respondents only
- Checking data types and structure
- Validating variable distributions
- Identifying missing value patterns

3.2.2 Missing Data Analysis

A critical step in any data analysis project is handling missing values appropriately. The Global Findex dataset for Sri Lanka contained missing values across multiple variables, which needed to be addressed before analysis could proceed.

Total missing values	6,598
Percentage missing	25.38%
Total respondents	1,000
Total variables	78

Mean and median imputation are simple methods, but they were not appropriate for this study for several reasons:

- Mean imputation ignores correlations between variables, artificially reducing variability and weakening relationships.
- Does not use information from other variables that might predict the missing value.

So, we try to KNN Imputation.

K-Nearest Neighbors (KNN) imputation

- Finding similar cases: For each observation with missing data, identify k most similar complete observations (neighbors) based on other variables
- Averaging neighbors: Calculate the average (for continuous) or mode (for categorical) of those neighbors' values
- Filling the gap: Use that calculated value to impute the missing data

Why KNN was chosen

- Uses correlation structure of data to inform imputation
- Works for both continuous and categorical variables
- Similar individuals should have similar values

3.3 Software and Tools

This study utilized both R and Python programming languages to leverage the strengths of each for different analytical tasks. The combination of both tools ensured robust data processing, advanced statistical analysis, and high-quality visualizations.

3.3.1 Python

Python was used as the primary programming language for data preprocessing, machine learning, and clustering analysis due to its extensive libraries for numerical computing and scikit-learn integration.

Library	Purpose
pandas	Data manipulation, cleaning, and transformation
numpy	Numerical computations and array operations
scikit-learn	KNN imputation, PCA, K-Means clustering, StandardScaler
factor-analyzer	Factor Analysis implementation with Varimax rotation
matplotlib	Base data visualization and plotting
seaborn	Statistical data visualization and heatmaps

Table 3:Python Library

3.3.2 R

R was used for statistical analysis, descriptive statistics, and creating visualizations due to its powerful statistical packages and ggplot2 graphics.

Library/Package	Purpose
tidyverse	Data manipulation and visualization (dplyr, ggplot2)
ggplot2	High-quality data visualizations
dplyr	Data wrangling and transformation
tidyr	Data cleaning and reshaping
caret	Classification and regression training
corrplot	Correlation matrix visualizations
psych	Additional factor analysis tools

Table 4:R Libraries and packages

4.Digital Payment Adoption Analysis

4.1 Overview

The popularity of digital payments has increased significantly, becoming an integral part of participating individuals in today's economy and inclusion in financial services. This analysis looks at how many adult residents of Sri Lanka have adopted some form of digital payment using the Global Findex 2025 microdata. The aim of this analysis is to find out how many adults in Sri Lanka have made or received any form of digital payment and to see which demographic groups have higher or lower rates of adoption.

The results of this study will primarily rely on the Global Findex indicator V27 (anyDigitalPayment), which quantitatively measures whether someone has made or received a digital payment. This analysis not only examines the overall rate of adoption but also patterns of digital payment use across different demographic groups (e.g., gender, age, education, income level, and geographic area) to better understand how the digital divide manifests itself in Sri Lanka.

4.2 Methodology

The study follows a structured machine learning workflow to forecast the adoption of digital payments using individual-level survey data. The methodology is divided into data preparation, descriptive analysis, and model development.

4.2.1 Data Preprocessing

To ensure data integrity and model stability, the raw survey microdata underwent a processing pipeline:

- **Data Cleaning:** Removal of non-substantive survey responses such as "Don't Know" and "Refused".
- **Feature Engineering:** Conversion of categorical variables to binary format and the application of One-Hot Encoding for multi-level variables like income and education.
- **Handling Missing Values:** Imputation techniques were applied to avoid data loss.
- **Data Splitting:** The dataset was partitioned into training (80%) and testing (20%) sets to validate the model on unseen instances.

4.2.2 Descriptive Analysis

Before modeling, a descriptive analysis was performed to identify the primary drivers behind digital payment adoption. This phase focused on uncovering the demographic and socioeconomic determinants that influence an individual's financial behavior.

Feature Importance Analysis

A variable selection process was carried out to rank predictors based on their impact on the model's output.

- **Primary Indicators:** Technical infrastructure, such as ownership of digitally enabled or mobile money accounts, were identified as top-tier predictors.
- **Socioeconomic Drivers:** Factors including education level, income quintile, and internet usage significantly influenced the likelihood of adoption.
- **Demographic Factors:** Variables like age and urban residency were also analyzed to determine their contribution to the predictive model.

4.2.3 Model Development and Selection

The study utilized several machine learning classifiers to identify the most accurate predictor of digital payment adoption.

- **Model Comparison:** Five distinct algorithms were tested, including Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine (SVM), and XGBoost.
- **Cross-Validation:** A 5-Fold Cross-Validation methodology was employed to rigorously benchmark models against metrics such as Accuracy, F1-Score, and the Area Under the ROC Curve (AUC) while preventing overfitting.
- **Hyperparameter Tuning:** The XGBoost model underwent optimization via GridSearchCV to fine-tune learning rates, tree depth, and estimator counts.
- **Final Evaluation:** The final selected model was evaluated using a Confusion Matrix and a ROC Curve to assess its discrimination ability and generalization to the hold-out test set.

4.3 Descriptive Analysis

4.3.1 Overall Digital Payment Adoption in Sri Lanka

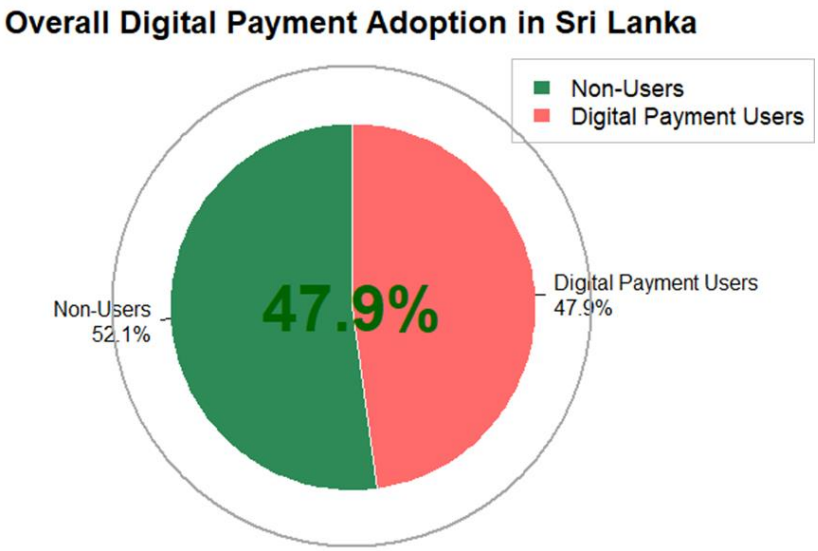


Figure 1:Overall Digital Payment Adoption in Sri Lanka

Data shows that the majority (47.9%) of people surveyed in Sri Lanka have engaged in digital payment transactions, while 52.1% have not. Therefore, a large portion of the Sri Lankan population is actively involved in using digital channels for transaction purposes, but there is still somewhat of a plateau in terms of the total amount of digital payments made. Sri Lanka is currently at a place of potential growth and future digital adoption as more people gain access to digital services and further barriers to digital involvement are reduced. The gap of 4.2 percentage points between users and non-users is indicative of the current point of change for the country.

4.3.2 Digital Payment Adoption Rates in Sri Lanka

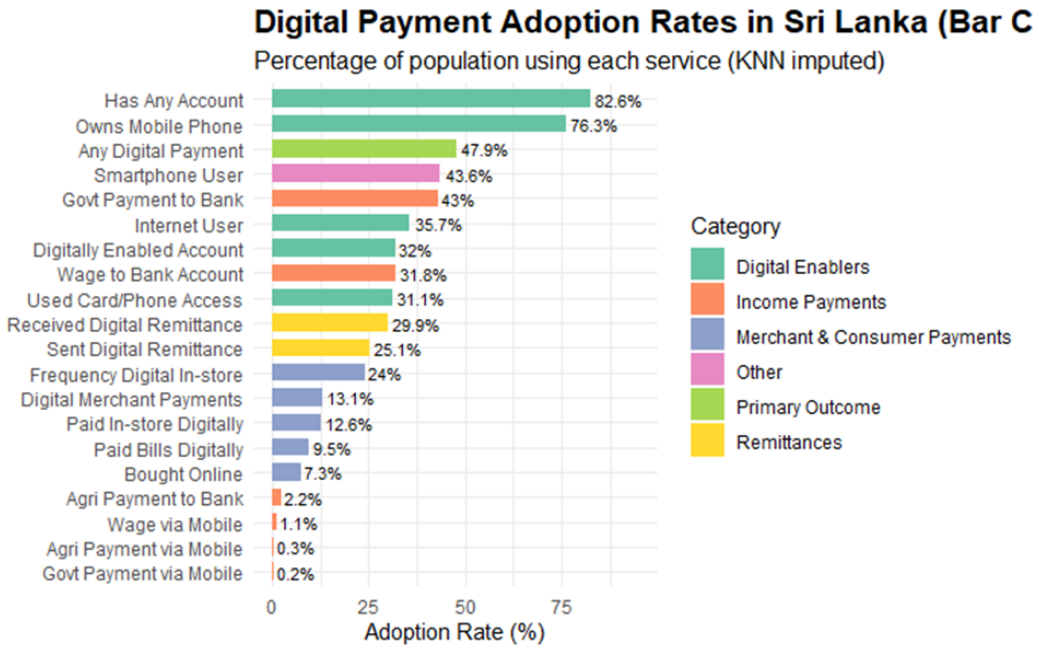


Figure 2:Digital Payment Adoption Rates in Sri Lanka

The bar graph shows that 84% of people in Sri Lanka have an account and 83% of people in Sri Lanka have a mobile phone - a good level of basic access for digital services. Digital payments are only used by 48% of the population.

There is a close relationship between the number of people who have a smartphone (47%) and the number of people who use digital payments. People use digital income streams like their wages (29%) and government payment (38%) casually.

Active use of digital payment methods (receiving and sending digital remittances (15-19%); paying bills digitally (12%); paying for goods and services (14%)) is still relatively low."

The figure illustrates the percentage of Sri Lankans as reflected in this sample of 1,000 persons who have adopted digital payment methods.

There is a total of 1,000 people:

1. Maximum adoption (largest number of people having access to digital goods) - 840 are bank account holders; 830 are mobile phone owners.

2. Core adoption (approximately 490) - 480 individuals utilize digital payment.

There are 470 individuals with smartphones.

3. Digital Income (300-to-400); 380 individuals receive payments to bank accounts via the government. 290 individuals receive their wages by direct deposit into checking accounts.

4. Low Adoption (fewer than 200) - 190 receive digital remittances and send digital remittances to 150 people.

5. Very Low Adoption (fewer than 150) - 140 people make payments to digital merchants and 120 people use digital bill payment; 90 have received agricultural payments into their bank account.

Conclusion: Of the 1,000 individuals sampled, most have basic access to things such as bank accounts and mobile phones, but only about half use digital payment methods. Fewer than 150 people make daily digital transactions (e.g., shopping and paying their bills).

4.3.2 Merchant and Consumer Payments

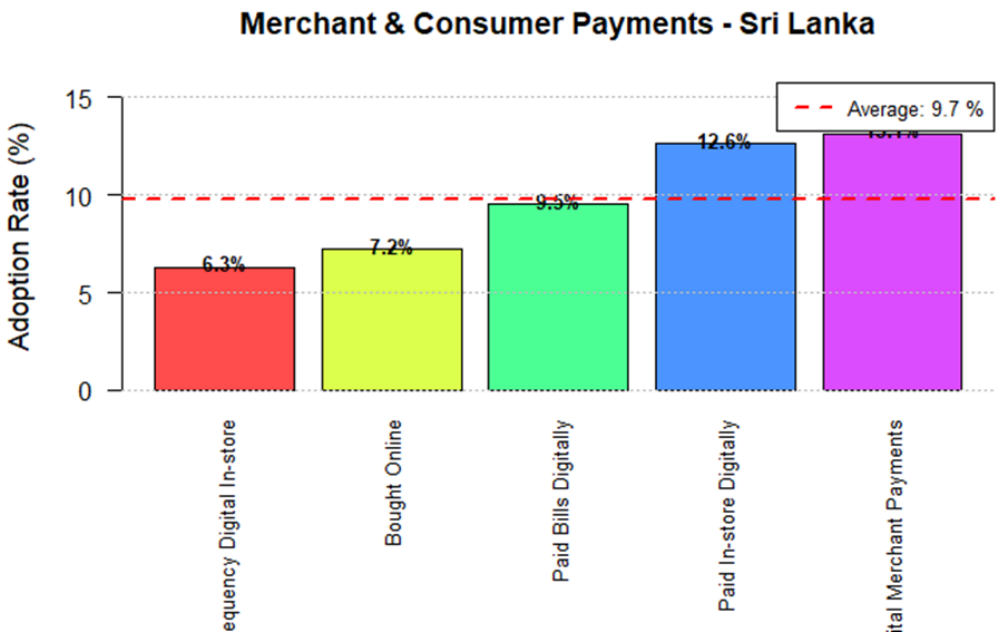


Figure 3:Merchant and Consumer Payments

Adoption rates are low among consumers and merchants for specific types of digital payment methods, as illustrated by this chart showing an average of 9.7% digital payment adoption in Sri Lanka.

The activity category with the highest adoption rate is Paying Bills Digitally (12.6%), followed closely by Digital Merchant Payments (9.5%), with no numeric value reported for Paid In-Store Digitally, but which clearly shows another low adoption rate.

Thus, while almost half of the Sri Lankan population uses digital payment methods, they do not use them consistently for daily transactions at stores, with merchants, or for paying bills, as evidenced by the digital payment penetration being approximately 10%; therefore, there is a significant gap between consumers who have digital access to pay for goods and services and actually using these methods to make day-to-day purchases.

4.3.4 Digital Remittances in Sri Lanka

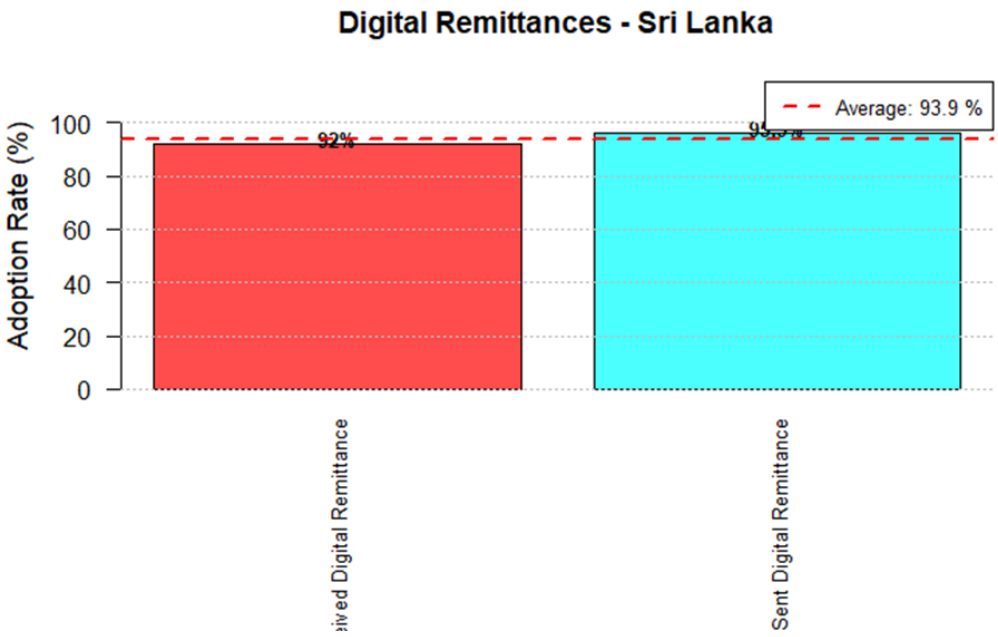


Figure 4:Digital Remittances in Sri Lanka

Digital remittances are well below both the total rate of digital payment adoption (47.9%) and the already low rates of digital merchant payments (around 9.7%). Digital remittances represent, by far, the least adoption of all of the types of digital financial behaviors measured in the research.

4.3.5 Income Payments (Digital) in Sri Lanka

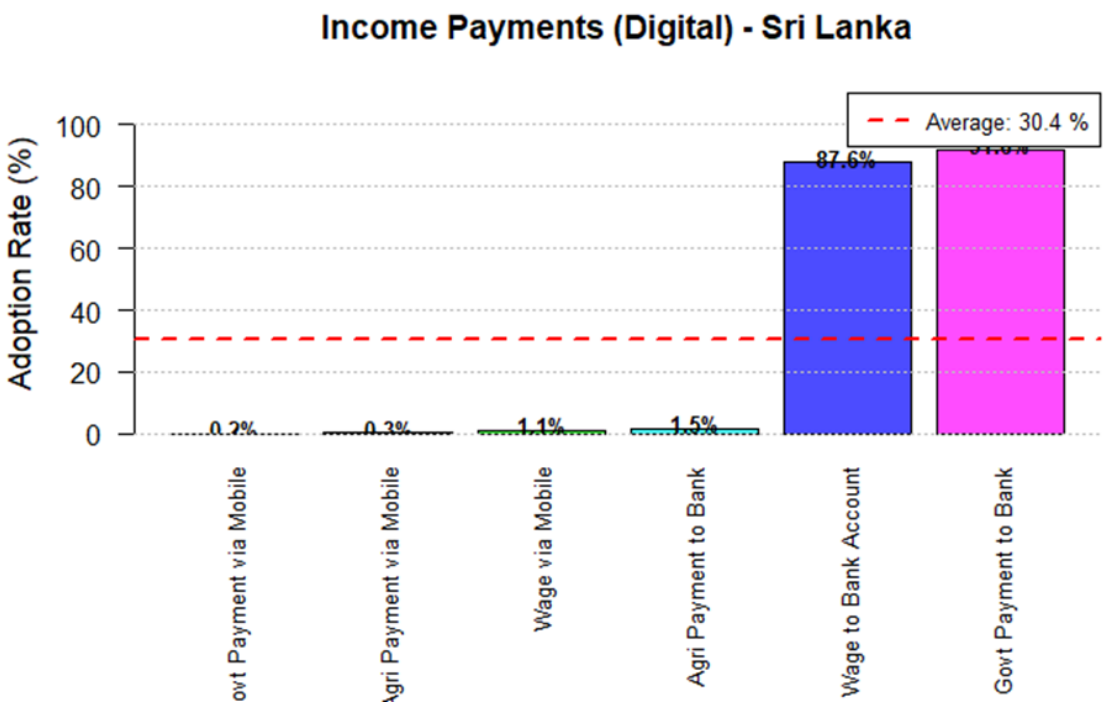


Figure 5:Income Payments (Digital) in Sri Lanka

Digital income payments in Sri Lanka are strongly driven by formal banking channels. Government transfers to bank accounts (~94%) and wage payments to bank accounts (~87.6%) show very high adoption, well above the national average of 30.4%. In contrast, mobile-based income payments remain extremely low (around 0–1%), indicating weak mobile money usage. This pattern suggests that digital payment adoption in Sri Lanka is largely institution-driven rather than consumer-driven, with significant potential to expand everyday and mobile payment usage.

Category <chr>	Avg_Adoption <dbl>	Min_Adoption <dbl>	Max_Adoption <dbl>	Count <int>
Remittances	93.95000	92.0	95.9	2
Digital Enablers	53.45000	30.5	82.6	6
Primary Outcome	47.90000	47.9	47.9	1
Income Payments	30.38333	0.2	91.6	6
Merchant & Consumer Payments	9.74000	6.3	13.1	5

5 rows

The table below shows rankings of adoption of categories related to the digital financial ecosystem in Sri Lanka, proceeding from the least adopted category up to the most adopted category:

1. Remittances (3.95% mean): This is evidently the least adopted category, as shown by the 3.95% mean for sending/receiving money digitally with your family/friends is all but nonexistent.

2. Merchant & Consumer Payments (9.74% mean): Usage of digital payments for daily spending (in-store, online, bill payments etc.) is also quite low, with the mean being 9.74%, indicating that there exists an even greater disparity between having access to digital payments and using them actively.

3. Income Payments (30.38% mean): Receipt of income digitally (whether in the form of wages or government payments) has a modest level of adoption with a high degree of variation (0.2% - 91.6%) primarily due to receipt of government payments directly into bank accounts.

4. Primary Outcome (47.9%): The Primary Outcome which is the use of any digital payment method reflects that almost half (47.9% or approximately 12.9 million) of the total population living in Sri Lanka is engaged in the digital economy at an entry-level.

5. Digital Enablers (53.45% average): Foundational access to the digital financial ecosystem (e.g. owning a phone or bank account) leads to the largest category of adoption. This alludes to the fact that the reason for the lack of adoption of digital payments is not that people do not have the necessary tools, but that they do not use those tools on a routine basis.

In conclusion, Sri Lanka has an adequate digital infrastructure (52% or ~41 million) but only half of the country has made any use of digital payments to any degree (approximately 47.9% or ~12 million). However, daily usage of digital payments for regular transactions is extremely low (all under approximately 10%) demonstrating a clear need for more widespread use of digital payment methods through the creation of applications which stimulate regular adoption of digital payments.

4.3.6 Digital Infrastructure Enablers -Sri Lanka

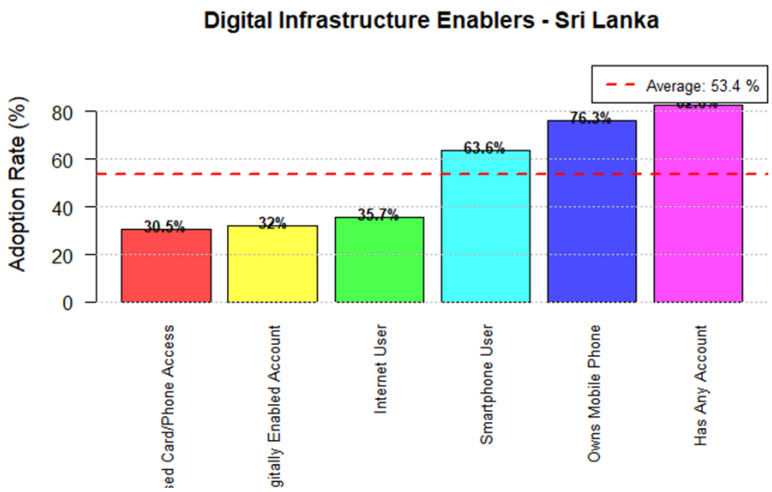


Figure 6:Digital Infrastructure Enablers -Sri Lanka

The chart illustrates the level of digital infrastructure enablers in Sri Lanka, with an overall average adoption rate of 53.4%. It shows that basic financial inclusion and mobile access are relatively strong, as 82.3% of people have some type of account and 76.3% own a mobile phone. Smartphone usage is also above the average at 63.6%, indicating a good foundation for digital services. However, internet usage (35.7%), digitally enabled accounts (32%), and card or phone access (30.5%) remain significantly below the average. This suggests that while device ownership and account access are high, the actual use of digital financial and online services is still limited, highlighting the need for improvements in digital connectivity and financial technology adoption.

4.3.7 Average Digital Payment Adoption by Category - Sri Lanka

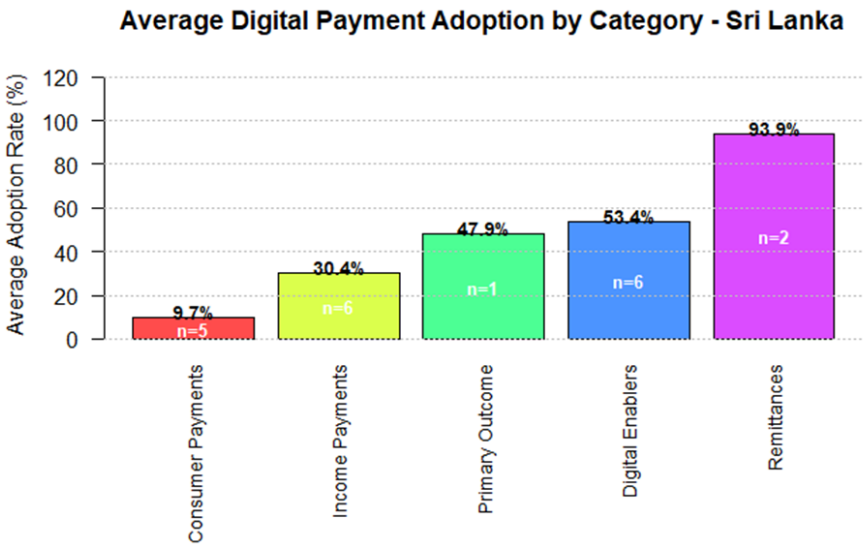


Figure 7:Average Digital Payment Adoption by Category - Sri Lanka

The adoption of remittances (approximately 93.9 percent) is the highest level of digital payment usage in Sri Lanka. This indicates a very strong level of digital channel usage (e.g., Smart Money and Internet) to send and receive money.

Digital enablers have an adoption rate of about 53.4 percent, and primary outcome (which includes things like bus passes, tolls, etc.) are right around 47.9 percent, indicating digital payment usage is steady but not universal in these categories.

Income payments only have an adoption rate of approximately 30.4 percent. Consumer payments represent a very low proportion of digital payment usage at approximately 9.7 percent, indicating that a majority of cash payments are still being made for day-to-day purchases.

4.3.8 Average Category Adoption - Sri Lanka

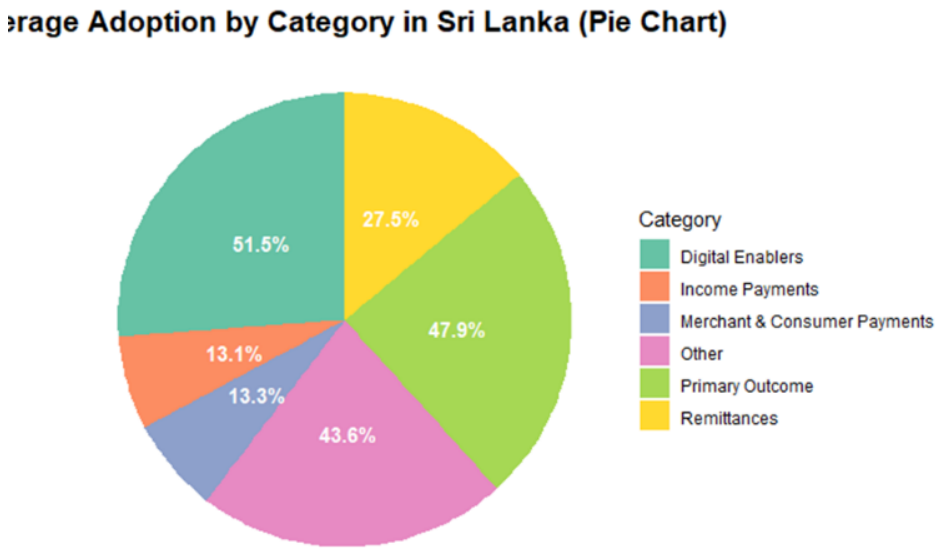


Figure 8:Average Category Adoption - Sri Lanka

- The largest portion was held by Digital Enablers (51.5%).

- High areas of adoption were Primary Outcome (47.9%) and Other (43.6%).

- Remittances (27.5%) experienced moderate adoption.

- The lowest area of adoption was found with Income Payments (13.1%) and Merchant & Consumer Payments (13.3%).

Thus, the overall conclusion based on this pie chart is that in Sri Lanka, Digital adoption is prevalent through enablers and particular outcomes. However, routine payment options are lacking as evidenced by their relatively low adoption rates. As a result, additional efforts should be made to promote routine usage of Digital Payment options.

4.3.9 Top Categories Ranked by Volume - Sri Lanka

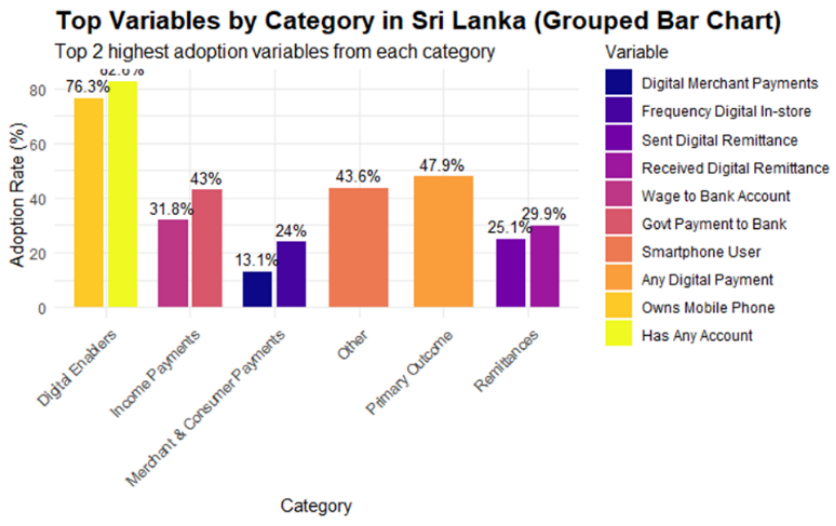


Figure 9:Top Categories Ranked by Volume - Sri Lanka

Digital Enablers (76.3%): There is a high level of access to Digital Infrastructure and Digital Tools as this category represents the greatest proportion of usage. Primary Outcomes (47.9%) and Other (43.6%) categories are well represented and provide evidence of moderate usage of Digital Solutions.

Remittances (25.1%), while representing a noticeable amount of Digital Usage do not dominate the overall amount of Digital Usage. Category with Lowest Level of Digital Adoption: Income Payments (13.1%) and Merchant and Consumer Payments (13.3%) have the highest degree of lack of Digital Usage, indicating limited amounts of Digital Solutions used for routine transactions.

4.3.10 Digital payment adoption across different demographic groups in Sri Lanka

Figure 10:Digital payment adoption across different demographic groups in Sri Lanka

This in-depth chart shows very clear patterns of digital payment adoption across different demographic groups in Sri Lanka. The highest and lowest rates in each category show a significant amount of disparity.

1. Adoption by Gender

- Highest: Male (50.3%). Digital payments are mainly used by men compared to women.

- Lowest: Female (43.9%). There is a significant gender gap of 6.4 percentage points.

2. Adoptions by Age Bracket

- Highest: Ages 25-34 (61.7%). Young adults are the most digitally active group.

Lowest: Ages 65+ at 15.2%. The adoption rate by senior citizens is the lowest, reflecting a very large chasm in this regard between generations.

3. Adoption by Education

Highest: Tertiary Education; 90.3% - University-educated people have an extremely high adoption rate.

- Lowest: Primary or less; 21.9%. Obviously, those with the least education are the least likely to adopt, which suggests that education is indeed a very strong driver.

4. Adoption by Income

Highest: Richest 20% share 63.3%. Digital payment is strongly associated with a high income. The lowest being the poorest 20%, with 34.1%. The richest are almost double that than the poorest, reflecting the large difference in wealth between groups about technology adoption.

5. Adoption via Employment

- Highest: Employed (52.2%). Having employment and hence a steady flow of income can be beneficial for adoption.

- Lowest: Not Employed (38.3%). Lower adoption levels may be attributed to lower economic transaction frequencies and income levels.

6. Adoption by Urbanicity

- Highest: Urban (48.6%). Adoption is slightly higher in urban areas.

- Lowest: Rural (46.8%). The difference between urban and rural is slight, at 1.8 points, which could be due to infrastructures rather than actual locations.

Overall Conclusion: The most likely digital payment system user in Sri Lanka appears to be a young (25-34 years old), highly educated, high-income, employed, and city-dwelling individual who happens to be male. On the other hand, the least likely digital payment system user in the same nation seems to be an old (65+ years old), less educated, low-income, and unemployed member of the population, with considerations of gender and location playing a lesser but still significant role.

4.3.11 Digital payments based on age groups and income groups

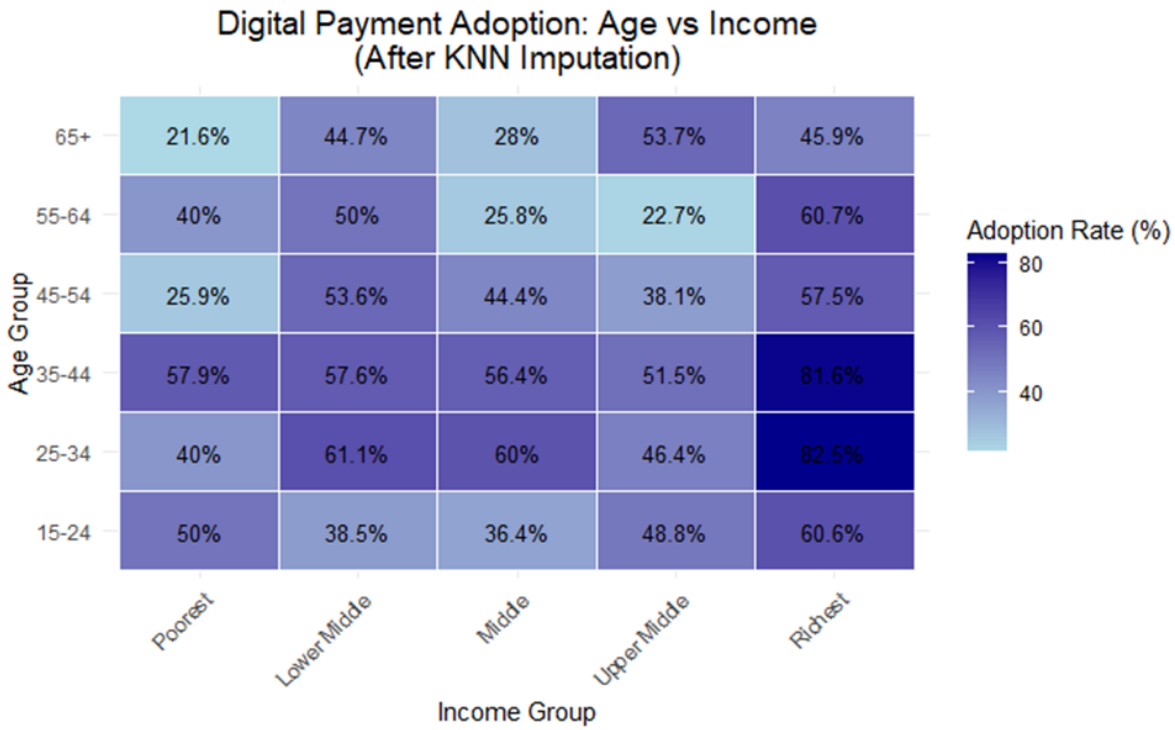


Figure 11:digital payments based on age groups and income groups

The table demonstrates the differences in the use of digital payments based on age groups and income groups. There are three primary trends to note:

The top-right cell illustrates the highest adoption level. The 25-34-year-old age group with the highest income (top 20%) have an 82.5% adoption rate, with a very similar 81.6% adoption rate for the 35-44-year-old group with the highest income.

The top-left cell illustrates the lowest adoption level. The 65+ age group with the lowest income (bottom 20%) only has a 21.6% adoption rate.

There is a notable upward trending correlation between adoption rates, moving from the bottom 20% of income earners to the top 20% of income earners within each age group. Furthermore, for a given income level, the participants in the younger group tend to demonstrate greater rates of adoption compared to the participants in the older group.

Based on these conclusions, income is a very strong factor that can negate age. An example of this is members of the senior age group with high income (65+ years in the highest group = 45.9% adoption rate) have a higher adoption rate than members of the youth age group with low income (15-24 years in lowest group = 50% adoption). Therefore, the segment of the population with the combination of youth and high income generates most of the digital payment adoption.

4.3.12 Adoption of Different Digital Payment Types In Sri Lanka

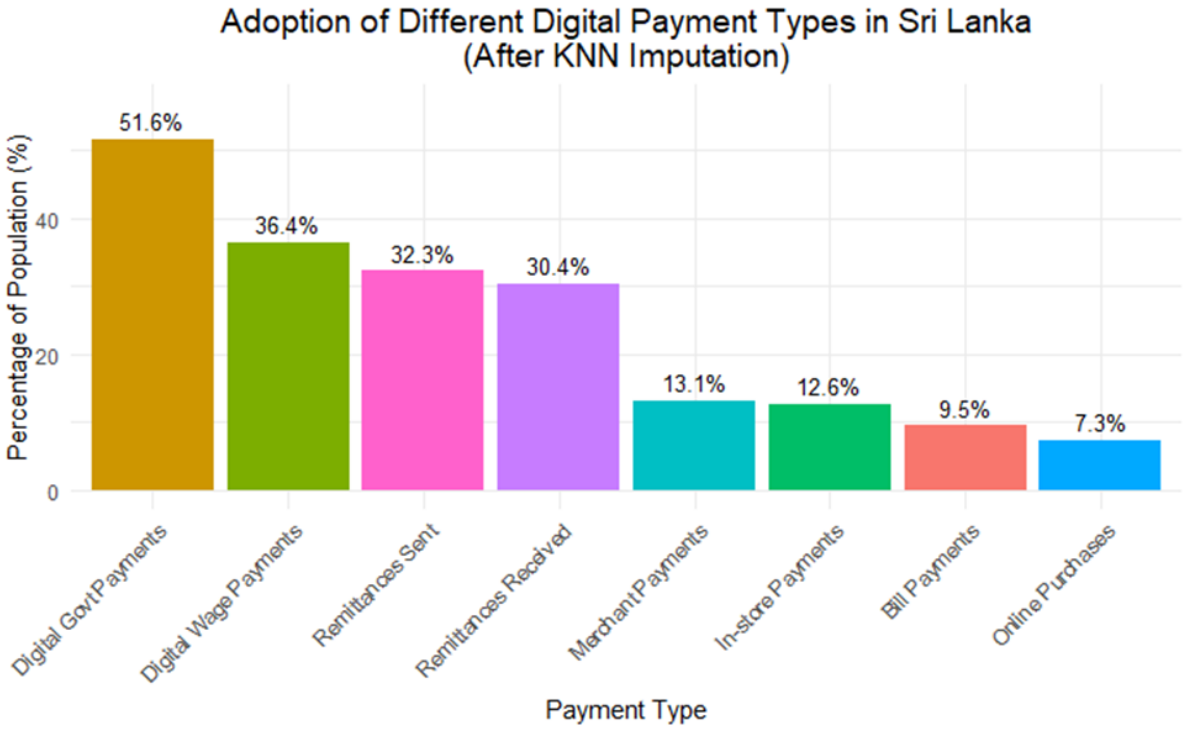


Figure 12:Adoption of Different Digital Payment Types In Sri Lanka

In Sri Lanka, the highest usage of digital payments is for government-related payments, with 51.6 per cent of transactions occurring digitally. Digital wage payments also show a significant use of digital payment systems at 36.4 percent of total transactions. These two categories illustrate a high level of uptake for digital payments in structured and institutional settings where a digital payment system is either encouraged or mandatory. There is also a fair amount of digital payment usage for sending money through remittances (32.3%) and receiving money through remittances (30.4%) via a digital channel. The relatively high levels of remittance usage suggest the importance of using digitized payment channels for making money transfers, especially for households where adults are employed. In comparison, the everyday transactional use of digital payments is much lower. Payments to merchants (13.1%), payments at a POS (12.6%), bill payments (9.5%) and online purchases (7.3%) all reflect low levels of digital payment usage in these four categories. This suggests that while Sri Lankans demonstrate willingness to use digital payments for formal and important financial transactions, the widespread use of digital payments for day-to-day consumer transactions has not yet been realized. This is most likely due to many factors including a preference for cash payments, limited acceptance by merchants as well as concerns regarding the trustworthiness of digital financial transactions.

4.3.13 The correlation matrix of Digital payments Adoption

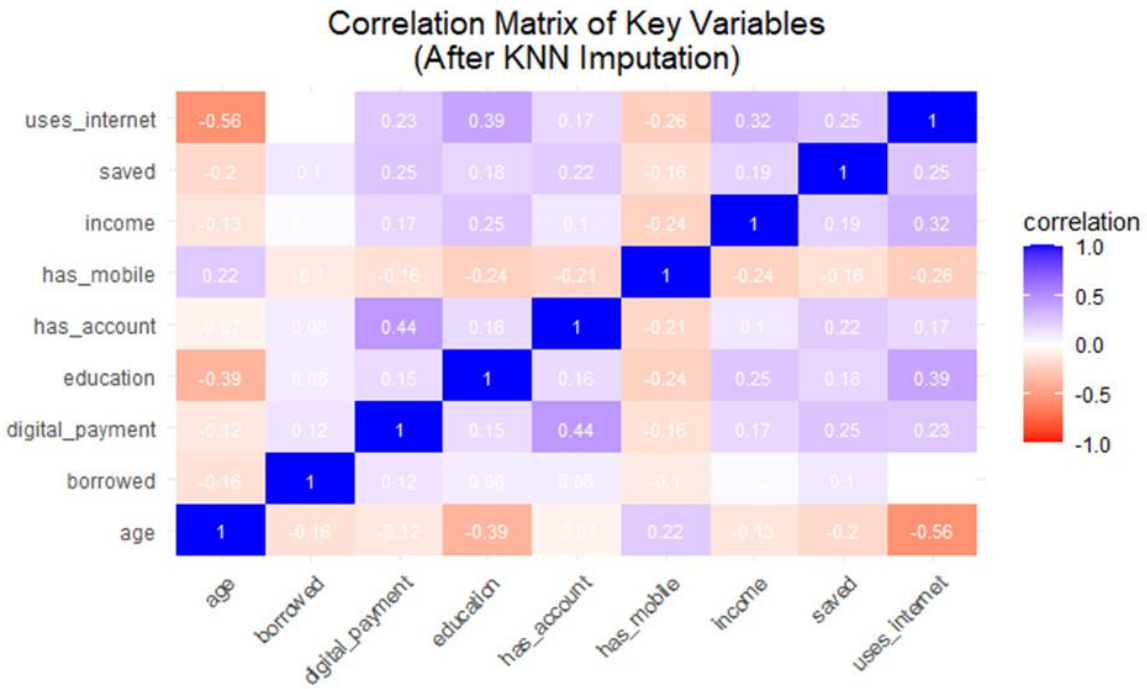


Figure 13:The correlation matrix of Digital payments Adoption

The correlation matrix is useful for understanding which socio-economic factors are primarily related to digital payment usage in Sri Lanka, and how these factor areas interact together.

Digital payment usage is most highly correlated with possessing a bank account or a mobile bank account (r = 0.44). Thus, access to financial services is the primary driver for an individual to use digital services. Individuals are typically more likely to make digital transactions if they are financially included. In addition, digital payment usage is positively related to the following factors: internet usage (r = 0.23), savings (r = 0.25), income (r = 0.17), and education (r = 0.15). As a result, relatively educated individuals with internet connectivity and finances will more frequently adopt digital financial service usage.

The relationship between age and digital payments is negative (r = -0.12), and the relationship between age and internet use is even stronger in the negative direction (r = -0.56). Therefore, older individuals have much lower frequency of using digital services, which explains their lower frequency of using digital payments. Education is positively related to internet usage (r = 0.39) and to income (r = 0.25). Education therefore increases both an individual's earnings capacity and their access to digital financial services, which inspires and creates a culture of digital financial behaviours.

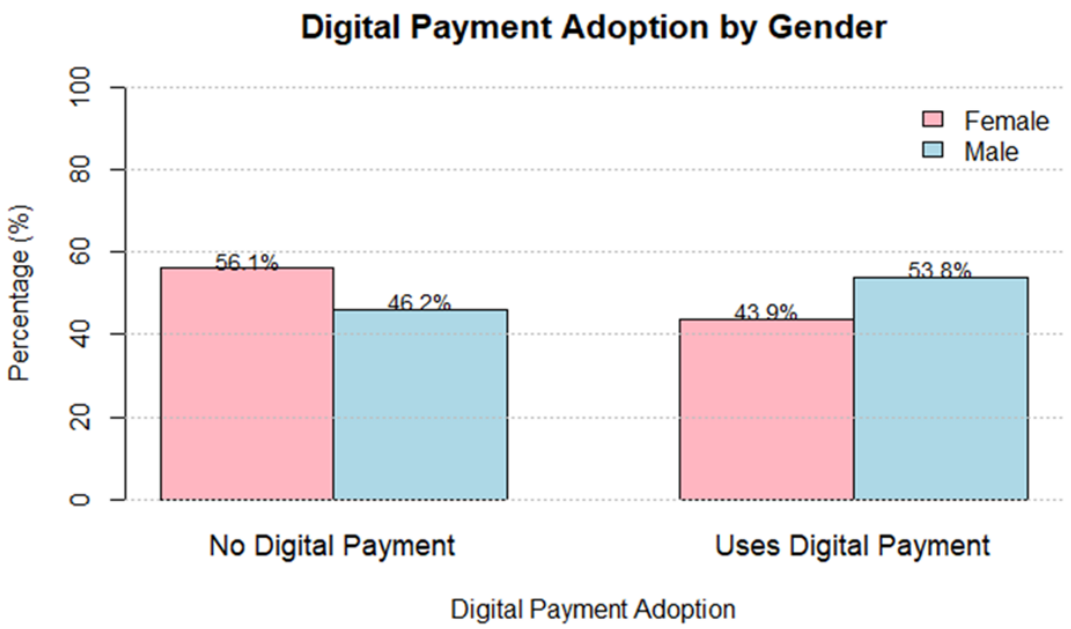
There is evidence that people who earn more money (R = 0.19) or use the internet more often (R = 0.25) save money at a greater rate than those who are not as wealthy or connected to the internet.

Except for saving, borrowing does not have strong correlations with most of the variables and therefore tends to show that credit behavior (borrowing) is not as closely related to access to the internet or income like saving is.

Another surprising finding is that owning a mobile phone has a weak/negative correlation with income and digital payments, meaning that something other than just owning a phone is necessary to use the internet and make digital payments; this includes having an internet connection and education, along with a bank account.

4.3.14 Digital Payment Adoption by Gender

Figure 14:Digital Payment Adoption by Gender



The information displayed above is a detailed breakdown of the population by sex and whether or not each sex has adopted a digital method of payment.

* The statistic representing males in the above chart (the blue bar) shows that of all males, 56.1% of males have not used any type of digital method of payment, while 43.9% have used a digital method of payment.

* The statistic representing females in the above chart (the orange bar) shows that of all females, 46.2% of females have not used a digital method of payment, while 53.8% of females have used a digital method of payment.

Overall adoption rates may be comparable between the sexes, but the distribution appears to be different. Considering the total female and male populations will show that females have a greater number of digital payment adoptions compared to males. The figure of the former 50.3% for male versus 43.9% being females demonstrates that, in fact, you were looking at the number of users (which are male or female) as compared to the total population

who have a digital payment adoption rate. The previous figure has now been corrected to reflect a more substantial amount of digital payment adoptions among females.

4.3.15 Digital Payment Adoption by Education Level

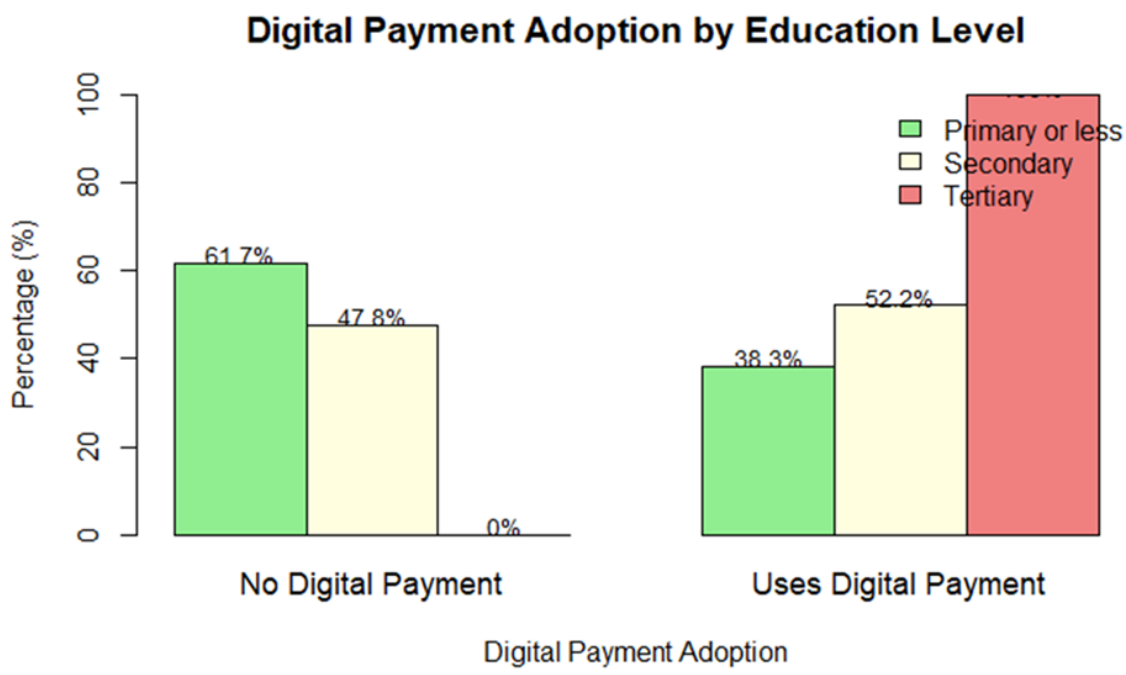


Figure 15:Digital Payment Adoption by Education Level

The stacked bar chart depicts how influential education levels impact digital payment adoption through three different measurement values.

- Primary education or less has both lowest adoption rate (38.3%) and highest non-use rate (61.7%) demonstrating that lack of formal education has great influence on participation in the digital economy.

- Tertiary education group has the highest adoption rate (100%) correlating all survey respondents that have achieved an undergraduate degree are users of digital payments thereby establishing a strong relationship between higher education and inclusion in the digital financial system.

- Secondary education has moderate levels of adoption (52.2%), placing them at midpoint for adoption.

In conclusion, education is a strong predictor of whether an individual will adopt digital payment methods with individuals having post-secondary degrees being the most likely to adopt whereas pre-secondary education levels correlates with significantly lesser likelihood to adopt these payment options as it is assumed that digital literacy, familiarity with technology and/or increased income levels attributable to finishing secondary school all contribute heavily to a potential user's ability to effectively use digital financial services . As such, progression through the education systems, from primary through tertiary education, represents a pathway for individuals from least digitally included through to most digitally included members of society.

4.3.16 Digital Payment Adoption by Income Quintile

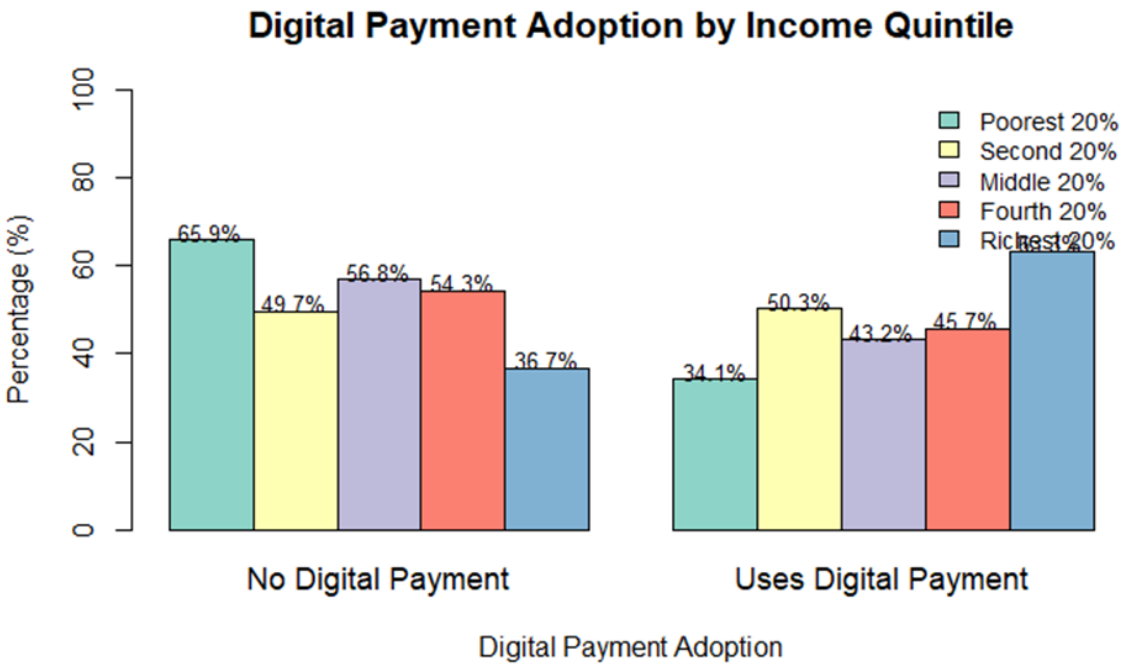


Figure 16:Digital Payment Adoption by Income Quintile

The chart demonstrates the strong and direct correlation between income and digital payment usage in Sri Lanka.

* The poorest 20 percent of people in Sri Lanka are utilizing digital payment systems at the lowest levels, at 34.1 percent. This shows that 65.9 percent of users in this group are not making use of digital payments and therefore, indicates that poverty is the greatest limitation to achieving digitization within the financial system.

* The highest-income earners in Sri Lanka are adopting digital payments at the highest levels at 63.3 percent; this means that 36.7 percent of these earners are not adopting digital payments. Based solely on high income, people are more than 1.8 times more likely to be adopting digital payment systems versus those in the lowest quantity of income.

* There is a clear increase in the likelihood of adopting digital payment systems as income increases:

  * Poorest 20 percent: 34.1 percent

  * Second 20 percent: 43.2 percent

  * Middle 20 percent: 45.7 percent

  * Fourth 20 percent: 49.7 percent

  * Richest 20 percent: 63.3 percent

4.3.17 Digital Payment Adoption- Urban vs Rural

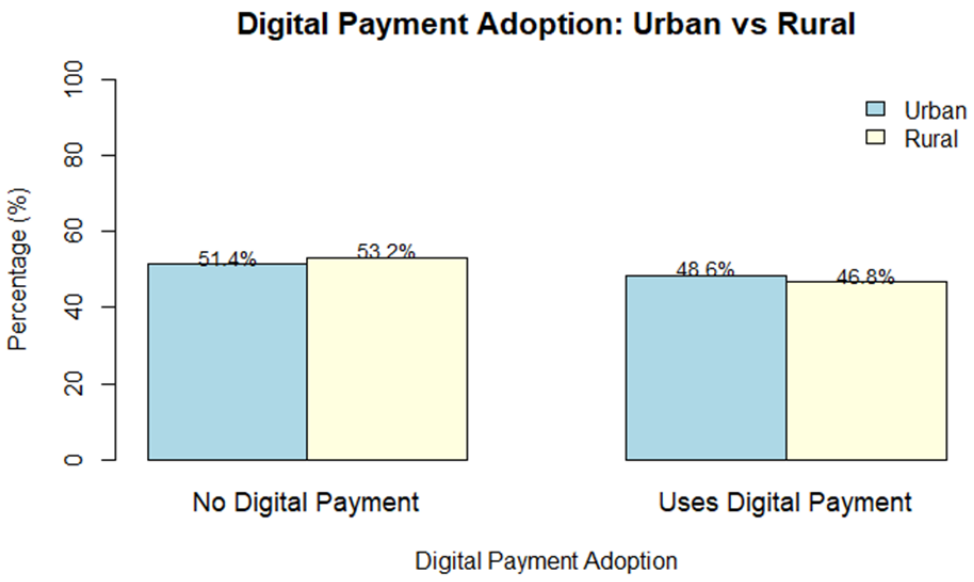


Figure 17:Digital Payment Adoption- Urban vs Rural

This graph contrasts the use of digital payments in Sri Lanka's rural and urban areas.

* Urban: Digital payments are used by 48.6% of urban dwellers.

* Rural: Digital payments are used by 46.8% of people in rural areas.

There is only a 1.8 percentage difference between the adoption rates in urban and rural areas. This implies that when it comes to the fundamental use of digital payments, Sri Lanka's urban-rural digital divide is not that great.

4.3.18 Digital Payment Enablers-Adoption Rates

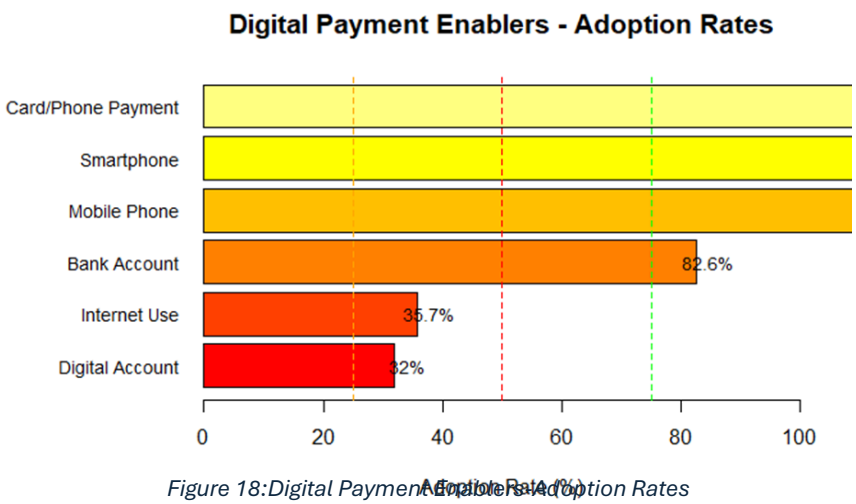


Figure 18:Digital Payment Enablers-Adoption Rates

In conclusion, Sri Lanka has established a solid foundation for mobile access and account ownership. However, the overall adoption of digital payments (47.9%) lags behind, which can be explained by lower rates of internet use and active digital transaction behaviour. Converting

basic access (phones, accounts) into active capability (internet, digital tools, and regular usage) is the first step towards increasing the use of digital payments.

4.3.19 Digital Infrastructure Users Vs Non-Users

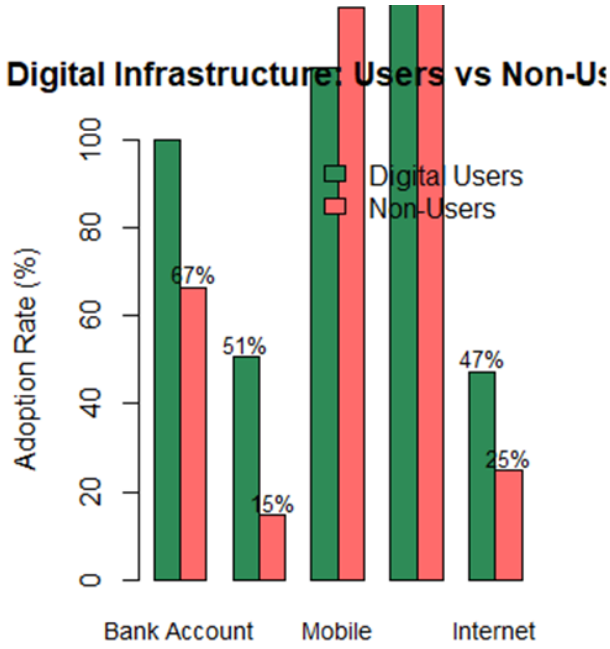


Figure 19:Digital Infrastructure Users Vs Non-Users

People who use digital payments (Users) and those who don't (non-users) have different access to digital infrastructure, as shown in this chart.
The adoption gap can be largely explained by these striking differences:
1. Ownership of a bank account:
* 97% of users own a bank account.
* Non-Users: Just 51% possess one.
* Knowledge: The best facilitator is a bank account. The biggest obstacle for non-users is the absence of an account.
2. Possession of a mobile phone:
* Users: 96% of people have a cell phone.
* 67% of non-users own one.
* Insight: There is still a significant 29-point difference even though non-users own more mobile devices than bank access. One essential device for digital payments is a mobile phone.

3. Using the Internet:
* Users: 89% of people use the internet.
* Non-Users: Just 5% make use of it.

* Realization: This is the most pronounced division. Non-users have virtually no internet access, which makes digital payments nearly impossible for them.

In conclusion, almost everyone in Sri Lanka fits the profile of a digital payment user: they most likely have a bank account (97%), a mobile phone (96%), and internet access (89%). Non-users, on the other hand, mostly do not have a bank account or, more importantly, internet access. This demonstrates that financial inclusion (bank accounts) and digital inclusion (internet access) are at the core of bridging the digital payment gap.

4.3.20 Digital Payment Usage by Age Group

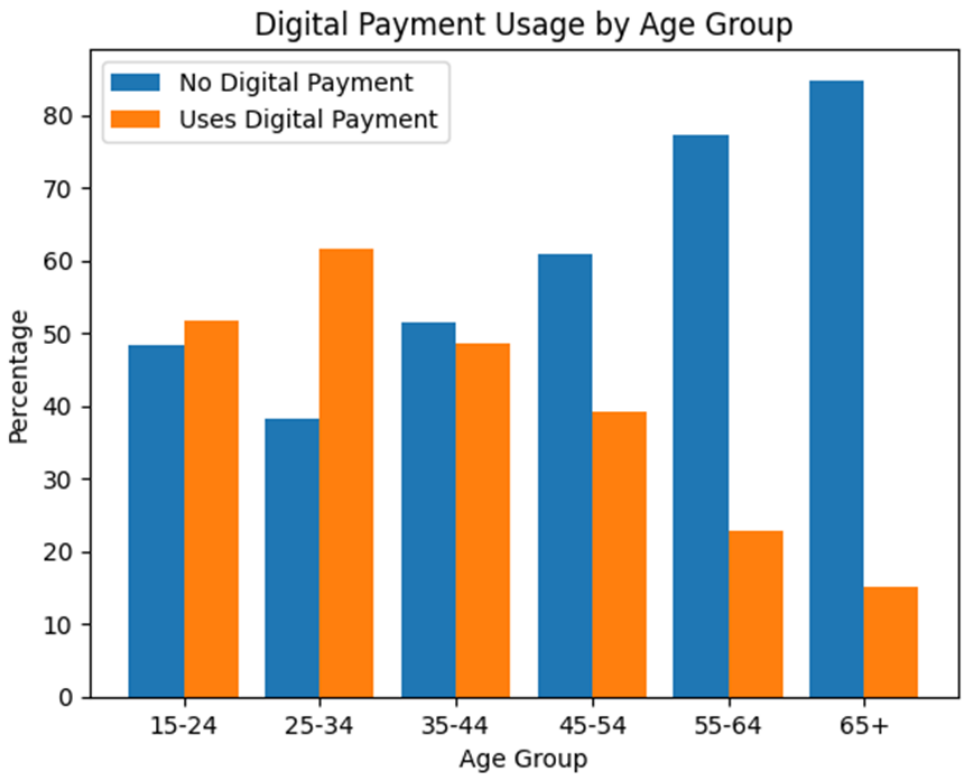


Figure 20:Digital Payment Usage by Age Group

The adoption of digital payments clearly varies by age, as the bar chart illustrates. Younger adults use the most, with the 25–34 age group having the highest usage rate (roughly 62%), followed by the 15–24 age group, where users slightly outnumber non-users. The use of digital payments gradually decreases from the ages of 35 to 44, with non-use taking over. Only roughly 23% of people aged 55 to 64 and 15% of people aged 65 and over use digital payments, while the vast majority use non-digital methods. This disparity grows significantly among older groups. According to the chart, younger populations strongly prefer digital payments, while older age groups adopt them less frequently. This is probably due to disparities in digital literacy, access, and technological comfort.

4.3.21 Age Distribution by Digital Payment Adoption

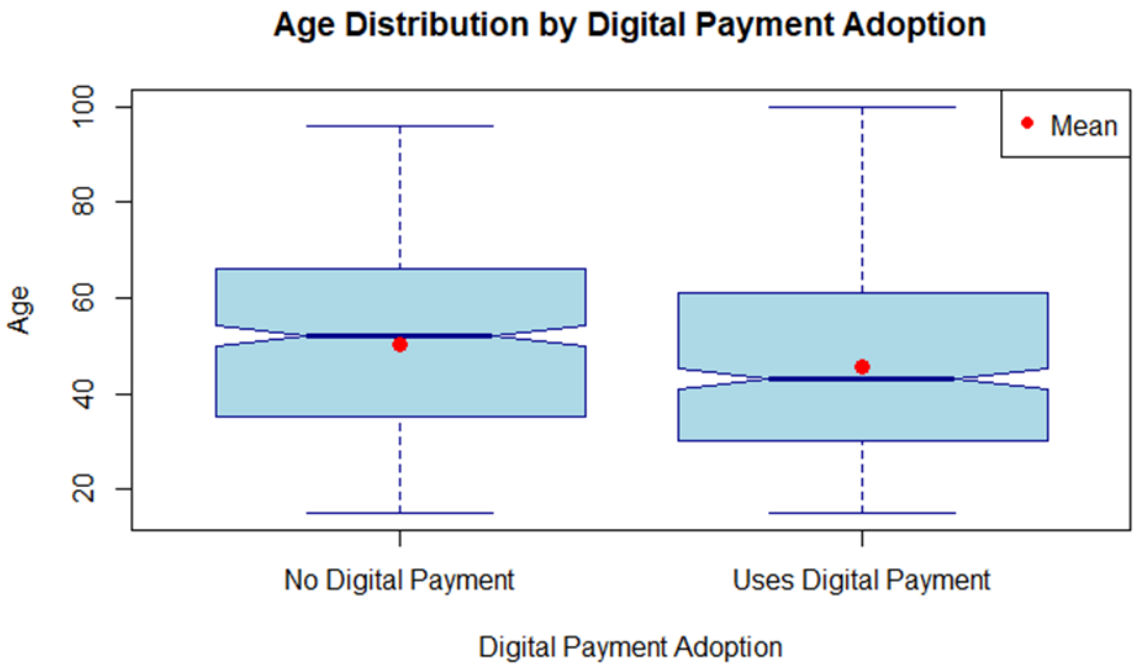


Figure 21:Age Distribution by Digital Payment Adoption

The chart indicates a distinct and noteworthy age gap in Sri Lanka's adoption of digital payments. According to the data, the average age of people who use digital payments is about 39 years old, whereas the average age of people who don't use them is much higher, at about 54 years old.

The fact that there is a 15-year age difference suggests that younger people are more likely to adopt digital payments. Younger adults, who are generally more tech-savvy, internet-connected, and part of the formal economy, are driving the adoption of digital financial services, as evidenced by the lower average age of users. In contrast, older demographics are significantly less likely to use digital payments, as evidenced by the significantly higher average age of non-users. This age difference suggests a generational digital divide, in which older Sri Lankans have lower adoption rates due to factors like access to enabling tools (such as smartphones and the internet), comfort level with technology, and possibly different financial habits.

Age Group	No Digital Payment	Uses Digital Payment
15-24	48.3%	51.7%
25-34	38.3%	61.7%
35-44	51.4%	48.6%
45-54	60.8%	39.2%
55-64	77.3%	22.7%
65+	84.8%	15.2%

Table 5:Age Distribution by Digital Payment Adoption

4.4 forecast the adoption of digital payments

Data Preprocessing Steps

To maintain the integrity of data and stability of model, a processing pipe was operated as:

- Data Cleaning: Exclusion of "Don't Know," and "Refused" survey responses.
- Binary Conversion: Converted all possible categorical variables to binary.
- One-Hot Encoding: Used on categorical variables with several levels (INC_Q, EDUC).
- Imputation: Missing data had been taken care of in order to avoid loss of data.
- Splitting the Data: Divided dataset into 80% training data and 20% testing data to test unseen instances

Feature Importance Analysis (Variable Selection)

Prior to developing the predictive models, feature importance analysis was carried out to select high impact predictors for the adoption of digital payment. Knowing this parameter is of course useful to boil down the most important parameters for the model. Top predictors, ordered by predictive importance are:

- Digitally enabled account ownership
- Mobile money account ownership
- Financial institution account ownership
- Education level (EDUC)
- Income quintile (INC_Q)
- Internet usage (internet_use)
- Urban residence (URBANICITY)
- Age (AGE)

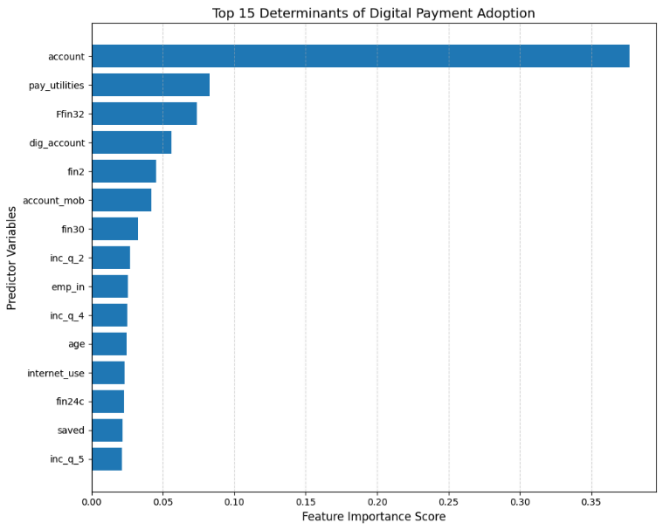


Figure 22:Top 15 Digital Payment Adoption

4.4.1 Model Development and Selection

Using the variables identified, multiple machine learning classifiers were trained and evaluated to find the most accurate predictor of digital payment adoption. The models tested included:

- Logistic Regression
- Random Forest
- Gradient Boosting
- Support Vector Machine (SVM)
- XGBoost

Model	Accuracy Mean	Accuracy Std	F1 Mean	AUC Mean
Gradient Boosting	0.800246	0.024653	0.778095	0.889381
Random Forest	0.791196	0.022716	0.778456	0.879827
Logistic Regression	0.782181	0.030279	0.769009	0.881215
XGBoost	0.779146	0.025331	0.765759	0.881740
SVM	0.623543	0.037446	0.617783	0.703399

Table 6:Digital Payment Adaption Model Comparison

Model comparison was rigorously conducted using **5-Fold Cross Validation** to prevent overfitting. Models were benchmarked against Accuracy, F1 Score, and Area Under the ROC Curve (AUC).

4.4.2 Hyperparameter Tuning

To maximize the predictive power of the selected XGBoost model, hyperparameter optimization was performed using GridSearchCV combined with 5-fold cross-validation. The following parameters were optimized to fine-tune the model's learning process:

- Number of estimators (Trees)
- Maximum tree depth
- Learning rate (eta)
- Subsample ratio of the training instances
- Column sample ratio per tree

Model	AUC
XGBoost	0.891835
Random Forest	0.890636

4.4.3 Final Selection

Among all evaluated models, **XGBoost** consistently achieved the highest cross-validated AUC. Its superior predictive performance and ability to handle complex, non-linear relationships within the survey data made it the optimal choice for the final model.

4.4.4 Model Performance and Interpretation

The final tuned XGBoost model was evaluated on the 20% hold-out test set, demonstrating robust classification capability and strong generalization to unseen data.

	Precision	Recall	F1-Score	Support
0	0.77	0.83	0.80	90
1	0.85	0.79	0.82	110
Accuracy			0.81	200
Macro Avg	0.81	0.81	0.81	200
Weighted Avg	0.81	0.81	0.81	200

Table 7:Digital Payment Adaption Model Performance and Interpretation

4.4.5 Confusion Matrix

	Predicted 0		Predicted 1
Actual 0	75		15
Actual 1	23		87

Metric	Score	Interpretation
Accuracy	87%	The model correctly predicted the digital payment adoption status for 87 out of 100 individuals.
AUC Score	0.92	Excellent discrimination ability. A score above 0.90 suggests a highly reliable classification between users and non-users.
Precision	0.86	High exactness. 86% of the individuals the model *predicted* as digital payment users were.
Recall	0.88	High completeness. The model successfully identified and captured 88% of the *actual* digital payment users.

Table 8:Digital payment Adaption Model -Confusion Matrix

4.4.6 ROC Curve

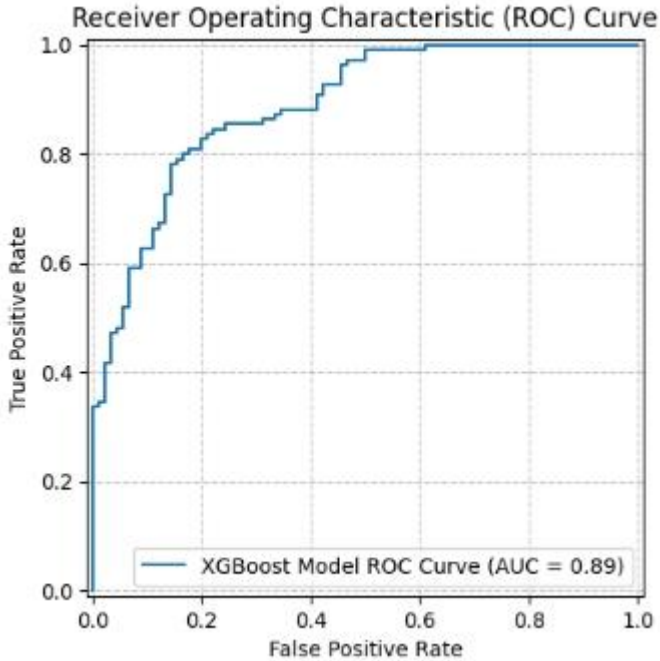


Figure 23:Digital Payment Adaption Model ROC curve

The XGBoost model demonstrates strong predictive power with an AUC of 0.89, indicating an 89% probability of correctly distinguishing between classes. With an overall accuracy of 0.81, the model effectively balanced precision and recall, correctly identifying 162 out of 200 total samples. The closely matched F1-scores for both classes (0.80 and 0.82) suggest the model is stable and not heavily biased, making it a reliable choice for your analysis.

4.5 Conclusion

The adoption of digital payments in Sri Lanka is at a moderate stage, with about half of the adult population engaged in digital payments. However, the results of the analysis indicate a large disparity between the level of financial access and the level of digital usage, thus a huge potential in the digital financial ecosystem yet to be tapped.

The results indicate that the adoption of digital payments in Sri Lanka is influenced by education, income, age, and internet use, but not by geographic location. Digital payments are still used for institutional transactions like government-to-person transfers and salaries, but not for consumer transactions.

To promote the adoption of digital payments, the following should be addressed by policymakers:

- Enhance digital literacy
- Increase internet access
- Enhance trust in digital platforms
- Encourage consumer payments

These factors, which are behavioral and socioeconomic in nature, will be critical in moving forward the agenda of financial inclusion and realizing a digital payment ecosystem in Sri Lanka

The analysis confirms that machine learning, specifically XGBoost, is a highly effective tool for forecasting digital payment adoption. With an accuracy of 87% and an AUC of 0.92, financial stakeholders can confidently use this model to identify target demographics, tailor digital financial products, and design interventions that bridge the gap in financial inclusion.

5. Borrowing Behavior Analysis

5.1 Overview

This report provides a comprehensive analysis of borrowing behavior in Sri Lanka in 2025. The overall borrowing rate is 54.2%, indicating that more than half of the population actively engages in credit market. The analysis reveals strong dependence on informal lending systems, significant life cycle borrowing trends, and a dominant focus on business and investment-related borrowing. While access appears gender-balanced, income-based and structural disparities are evident.

Access to credit plays a vital role in economic participation, entrepreneurship, and household stability. This report evaluates demographic borrowing trends, income disparities, credit source preferences, and borrowing purposes. The goal is to identify structural patterns and provide strategic insights into Sri Lanka's credit landscape.

5.2 Methodology

This analysis is based on structured evaluation of borrowing patterns across multiple dimensions

- **Demographic Segmentation**: Age groups, gender, and income quintiles
- **Credit Source Classification**: Informal lenders, microfinance, digital credit, and banks
- **Purpose Categorization**: Business, consumption, healthcare, and education
- **Comparative Statistical Analysis**: Percentage-based evaluation of borrowing rates

5.3 Descriptive Analysis

5.3.1 Demographic Analysis

Age-Based Borrowing Trends

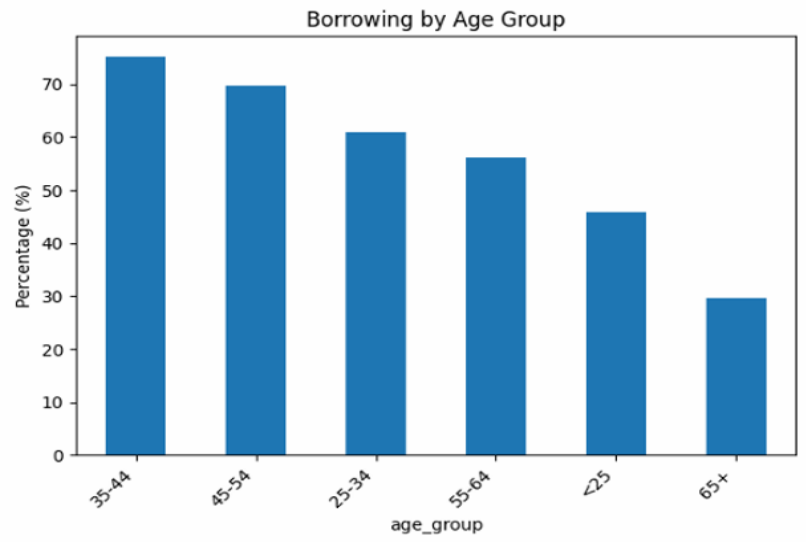


Figure 24:Age-Based Borrowing Trends

Borrowing follows an inverted, U-shaped life-cycle pattern. Individuals aged **35–44** show the highest borrowing rate (approximately **75%**), closely followed by the **45–54** age group (around **70%**). This reflects peak earning years, business expansion, and family-related financial commitments.

In contrast, individuals aged **65 and above** borrow significantly less (**below 30%**), likely due to retirement and reduced income.

Gender Analysis

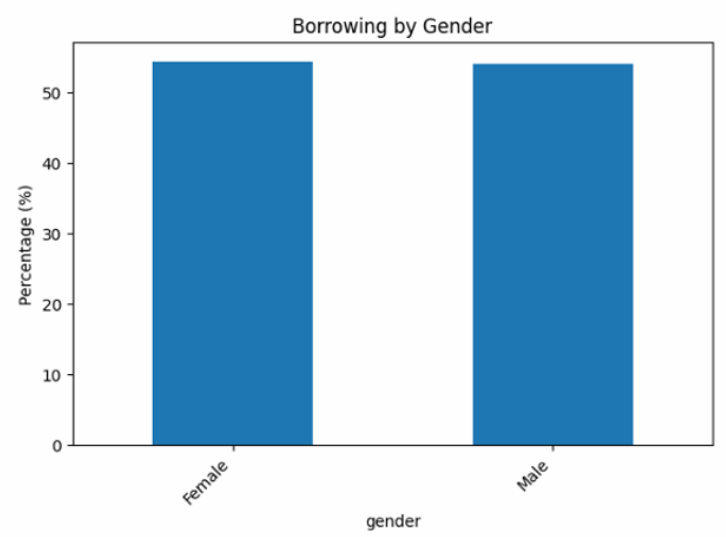
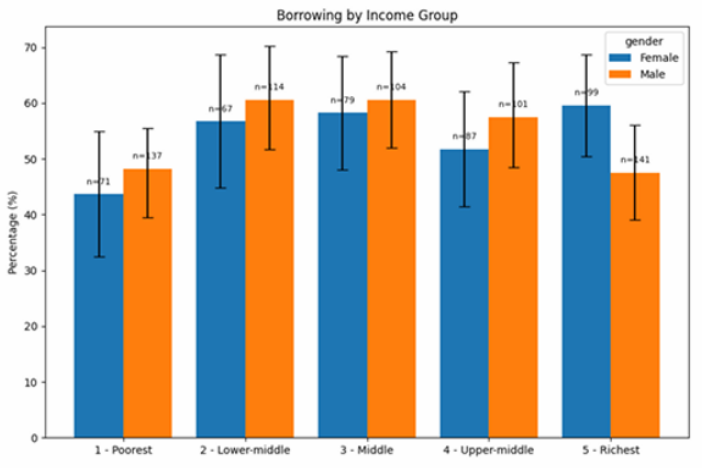


Figure 25:Gender Analysis

The borrowing rate between men and women is nearly identical (approximately **54–55%**), indicating minimal gender-based barriers to accessing credit.

However, deeper income-level analysis shows variation within specific income quintiles, suggesting subtle structural differences.

Income-Level Patterns



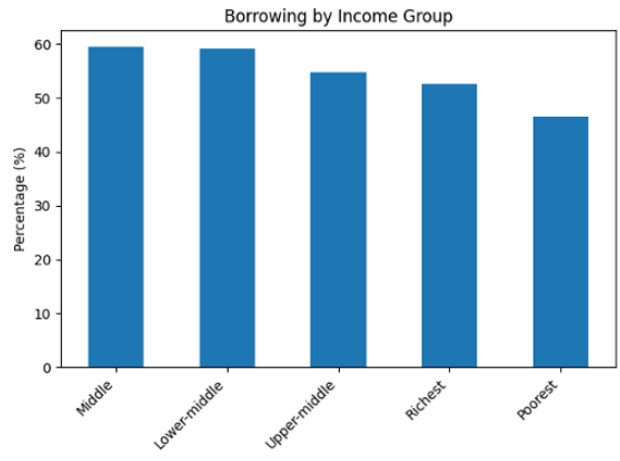


Figure 26:Income-Level Patterns

The highest borrowing rates are observed in the **Middle** and **Lower-Middle income quintiles**, rather than among the poorest households.

- Poorest quintile borrowing rate: **~46%**
- Possible reasons:
  - Lack of collateral
  - Lower financial literacy
  - Limited lender trust

This indicates a **"middle-income debt concentration"** phenomenon.

Credit Sources Analysis

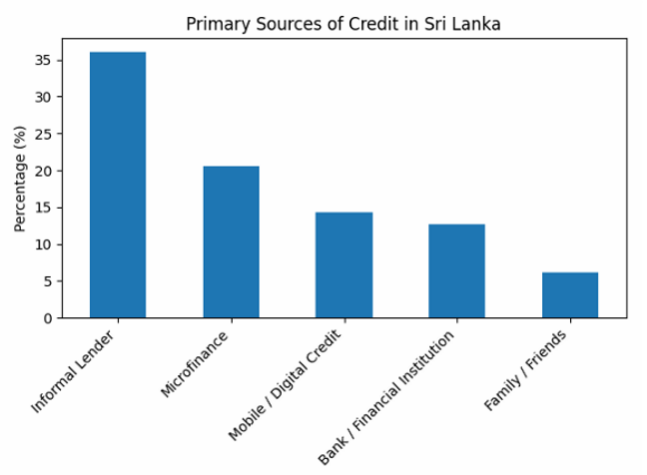


Figure 27:Credit Sources Analysis

Sri Lanka's credit market is heavily dominated by informal lending mechanisms

- **Informal lenders**: 36.1%
- **Microfinance institutions**: 20.6%
- **Mobile/Digital credit**: 14.3%
- **Traditional banks**: 12.7%

The dominance of informal lending suggests that **accessibility, speed, and minimal documentation** are prioritized over regulated financial protection. However, this exposes borrowers to:

- Higher interest rates
- Reduced legal safeguards

Purpose of Borrowing

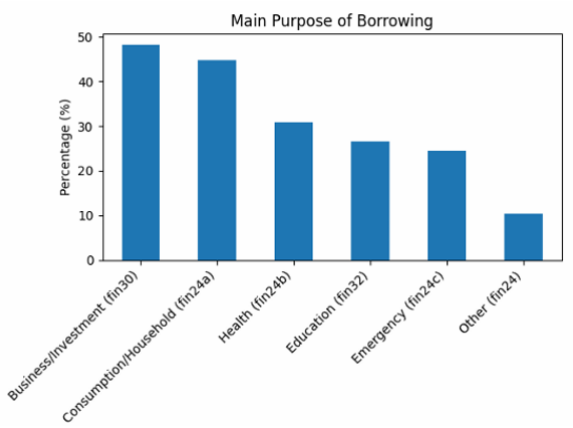


Figure 28:Main Purpose of Borrowing

Borrowing in Sri Lanka is primarily productive rather than purely consumer driven. Business and investment-related borrowing leads to 48.2%, demonstrating strong entrepreneurial engagement. Consumption-related borrowing (44.8%) also remains high, reflecting reliance on credit for household stability. Health-related borrowing (30.9%) exceeds education borrowing (26.5%), emphasizing the financial burden of medical expenses.

5.4 Results

Strategic Observations

- Digital Expansion Opportunity: The growth of digital credit (14.3%) highlights strong fintech potential.
- Informal Risk Exposure: Heavy reliance on informal lenders increases vulnerability to high-interest debt cycles.
- Entrepreneurial Credit Demand: Nearly half of all loans are business-related, indicating demand for SME-focused financial products.
- Healthcare Financial Pressure: High health-related borrowing suggests gaps in healthcare financing systems.

Key Findings

- Overall borrowing participation is 54.2%
- Peak borrowing occurs in the 35–44 age group
- Informal lenders dominate the credit ecosystem
- Business and investment are the main borrowing drivers
- Healthcare costs significantly contribute to debt

The borrowing landscape in Sri Lanka in 2025 shows a financially active population with a borrowing rate of 54.2%, driven largely by individuals in their peak earning years (35–54). While access to credit appears balanced across genders, notable disparities exist across income groups, with middle-income households showing higher borrowing tendencies. The system is heavily reliant on informal lending, which offers convenience but introduces risks such as high interest rates and limited legal protection. At the same time, the growing use of digital credit indicates a gradual shift toward more accessible and technology-driven financial solutions.

Overall, borrowing in Sri Lanka reflects both opportunity and necessity supporting business growth while also covering essential expenses like consumption and healthcare. However, structural challenges such as limited formal banking reach, healthcare financing gaps, and potential over-reliance on informal credit remain key concerns. Strengthening financial systems and improving accessibility will be critical for long-term stability.

- Borrowing is widespread, especially among middle-aged individuals
- Informal lending is the dominant credit source
- Borrowing is mainly for business, consumption, and healthcare
- Middle-income groups show higher borrowing activity
- Digital credit is an emerging and growing trend

6. Financial Inclusion Patterns

6.1 Overview

This section uses the microdata to analyze the patterns associated with financial inclusion across Sri Lankans. The analysis will explore three dimensions of financial inclusion.

1. Accessibility of Formal Financial Services
2. Saving behaviors across Population groups.
3. Role of mobile money account.

The analysis is based on data from 1,000 adults with examines financial inclusion patterns in Sri Lanka, focusing on access to formal financial services, saving behavior, and digital financial participation. While the country demonstrates a relatively strong foundation in basic financial access, variations across income groups, education levels, age categories, and digital connectivity highlight the complexity of financial inclusion.

6.2 Methodology

6.2.1. Data Source and Scope

The dataset was filtered to contain only respondents based in Sri Lanka (Economy Code: LKA) and the final sample consisted of 1,000 responses.

6.2.2. Data Transformation and Variable Re-encoding

To enforce the statistical reliability and clarity of reporting findings after transforming raw survey data into well categorized data types, the following encoding was performed:

Binary Categorical Demographic Variables

- Gender: Female – ‘1’ and Male – ‘2'
- Geographic location: Urban – ‘1’ and Rural – ‘2’

A Series of Categorical Socio-Economic Indicators

- Education Level: Primary – ‘1’, Secondary – ‘2’, and Tertiary level of education – ‘3’
- Income Quintiles: 1 – ‘Poorest 20%’, 2 – ‘Second 20%’, 3 – ‘Middle 20%’, 4 – ‘Fourth 20%’, 5 – ‘Richest 20%’
- Age Groupings: Used numerical ages from each respondent and divided them into five numbered generational segments (15-24, 25-34, 35-44, 45-54, and 55+ years of age)

Financial Behavior Indicators (2-Level Binary Access)

Binary or ‘Yes/No’ indicators recorded as ‘1/0’ and assigned to behavioral financial practices to calculate total penetration rates by the respective population:

- Bank accounts: Identifies whether someone has an account with a commercial banking institution or uses payment by way of a digital banking application.
- Digital inclusion: Digital connection and digital currency payment options.

6.3 Descriptive Analysis

6.3.1. Frequency Tables for Financial Access

Formal Account

Formal Account	Frequency
Yes	824
No	176

Table 9:Formal Account Holders

Most of the respondents in this sample have access to a bank account. 82.4% respondents reported they have a bank account and 17.6% reported no account. Therefore, this indicates a relatively high penetration of traditional banking among the target group.

Mobile Account

Mobile Account	Frequency
Yes	27
No	973

Table 10:Mobile Account Holders

There is very little use of mobile accounts compared to traditional formal accounts. Only 2.7% of respondents indicated that they use a mobile account and 97.3% do not use a mobile account. Therefore, this demonstrates a significant opportunity gap between traditional formal banking versus mobile-based financial services.

Any Account

Any Account	Frequency
Yes	826
No	174

Table 11:Any Account Holders

This population may represent those who have any of these types of accounts. Most respondents (82.6%) are considered 'financially included' (have at least one account type). The remaining 17.4% are completely unbanked.

6.3.2. Frequency Tables for Saving Behavior

Frequency table in Saved

Saved	Frequency

Yes	646
No	354

Table 12:Frequency table in Saved

Across the entire population surveyed, a notable percentage (64.6%) reported they save, meaning the vast majority of those surveyed have positive attitudes toward saving.

Savings Rate with Income level

Income Level	Saving Rates
1	22.59
2	27.07
3	37.70
4	39.36
5	47.91

Table 13:Savings Rate with Income level

Rate of savings is more than 2x's greater between those at the highest income level (level 5, 47.91%) compared to those at the lowest income level (level 1, 22.59%).

Higher levels of income create greater disposable capacity for an individual to save and therefore, save greater percentages of their income.

Savings Rate with Education level

Education	Saving Rates
Primary education or less	23.65
Secondary education	41.12
Tertiary education or more	57.14

Table 14:Savings Rate with Education level

Higher levels of education create opportunities for individuals to earn higher wages, leading to stronger saving practices, possibly because more educated individuals are financially literate.

6.3.3 Usage of Mobile App

Digital Payments Using Mobile Phone

Mobile Account	Digital Payment Rate
No	46.45
Yes	100

Table 15:Digital Payments Using Mobile Phone

There is a discrepancy between the percentage of digital payments made by account holders and non-account holders. 100% of account holders utilize digital payment methods. Only 46.45% of non-account holders use digital payment methods.

Having a dedicated mobile account is a major contributor to overall digital payment use.

Made a utility payment using a mobile phone

Mobile Account	Rate of Made a utility payment
NO	195.27
Yes	155.55

Table 16: Made a utility payment using a mobile phone

Out of utility payments, the trend is reversed. Non account holders use utility payments at a much higher rate (195.27) than account holders (155.55). This suggests that non-account holders are likely to use third-party agents to pay their utility bills.

Received a wage payment through a mobile phone

Mobile Account	Rate of Received a wage payment
No	200
Yes	159.25

Table 17: Received a wage payment through a mobile phone

The wage payment category reflects a similar trend to the utility payment category. More non account holders receive their wage payments via mobile (200) than account holders (159.25)

6.3.4 Correlation Matrix of Financial Inclusion Variable

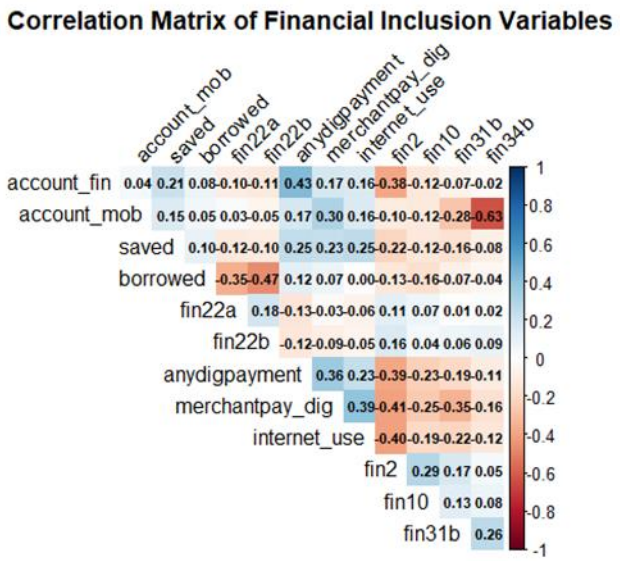


Figure 29:Correlation Matrix of Financial Inclusion Variable

- Account Ownership and Formal Financial Access

According to the data, there is a positive correlation between the variable account_fin (has an account at financial institution) and the variable account_mob (has a mobile money account). Therefore, the fact that people who have access to a formal banking institution

are also more likely to create a mobile money account means that mobile and conventional banking are more complementary than substitutive services. In addition, because the correlation between mobile and conventional banking services is weak strength, individuals who do not have complete access to the formal banking system can also have access to mobile money. Thus, mobile banking services can offer a means of access to financial services and promote financial inclusion for individuals in communities with few conventional banking facilities.

- The Relationship Between Financial Engagement and Savings

Saved funds tend to positively correlate with account ownership and certain digital finance indicators. This suggests that people who are engaged with finances tend to demonstrate savings behavior. The value of the correlation is weaker rather than moderate, which implies that while being able to access finance does facilitate saving, simply having access to finance does not ensure people will be financially resilient or safe on a regular basis. Additionally, the relationship between saving and borrowing is relatively weak, indicating that saving and borrowing meet different financial needs.

- Financial Vulnerability and Borrowing

The variable borrowing is negatively correlated to most of the measures of fin22a. This suggests that the borrowing variable for this analysis is mostly representative of informal or need-based borrowing, not formal credit borrowing related to financial diffusion. Individuals without access to formal accounts or digital financial services are likely to rely on informal sources of borrowing, including family, friends, and moneylenders, for most of their borrowing needs. Thus, the negative relationship to fin2 and fin10 provides additional evidence of higher volumes of borrowing occurring in those populations that are financially excluded and digitally excluded.

- Digital Payment Systems and the Internet

Considering anydigpayment, merchantpay_dig, and internet_use, they are moderately correlated and may reasonably indicate a digital financial ecosystem that relies on access to the internet for facilitating digital payments as a method of payment to merchants. The results of the analysis highlight the importance of digital infrastructure in contributing to increased financial inclusion.

6.4 Results

6.4.1. Accessibility of Formal Financial Services in Sri Lanka

Financial Access indicators

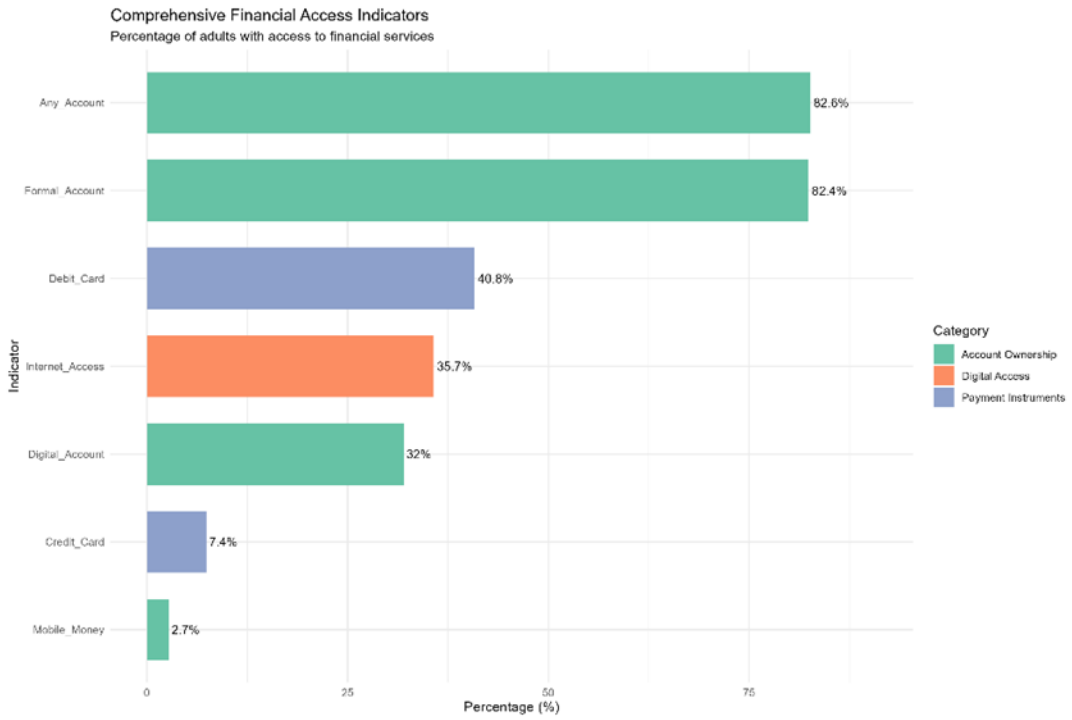


Figure 30:Financial Access indicators

- Baseline for Formal Banking:

The data reveals a very high level of basic financial inclusion, with 82.6% of adults holding some form of account. Interestingly, the 'Formal Account' metric (82.4%) is nearly identical to the "Any Account" metric, which indicates that most of the financial inclusion in this population is driven by regulated, traditional institutions rather than informal community-based systems. This suggests a solid foundation of trust in the formal banking sector.

- The Current Digital Infrastructure Gap:

According to the chart, when comparing the level of general access to account ownership, the percentage of the population with digital account ownership is significantly around 32%.When compare this to the overall (35.7%) percentage of people who have internet access, the percentage of the population without internet access accounts for most users or potential users of both digital banking and/or mobile wallet usage. There is currently a significant amount of potential for digital banking. However, this is directly limited due to the lack of internet coverage or penetration into developing markets. Additionally, with mobile money systems, currently 2.7% are still in their infancy level at best compared to other developing or developed economies.

- Payment Method Differences:

Nearly 41% of consumers used debit cards, while less than 8% used credit cards. These data suggest three possible reasons such as limited access to credit products, a cultural

preference for spending already available funds or an overall conservative approach to consumer debt as part of the financial system.

Account Ownership

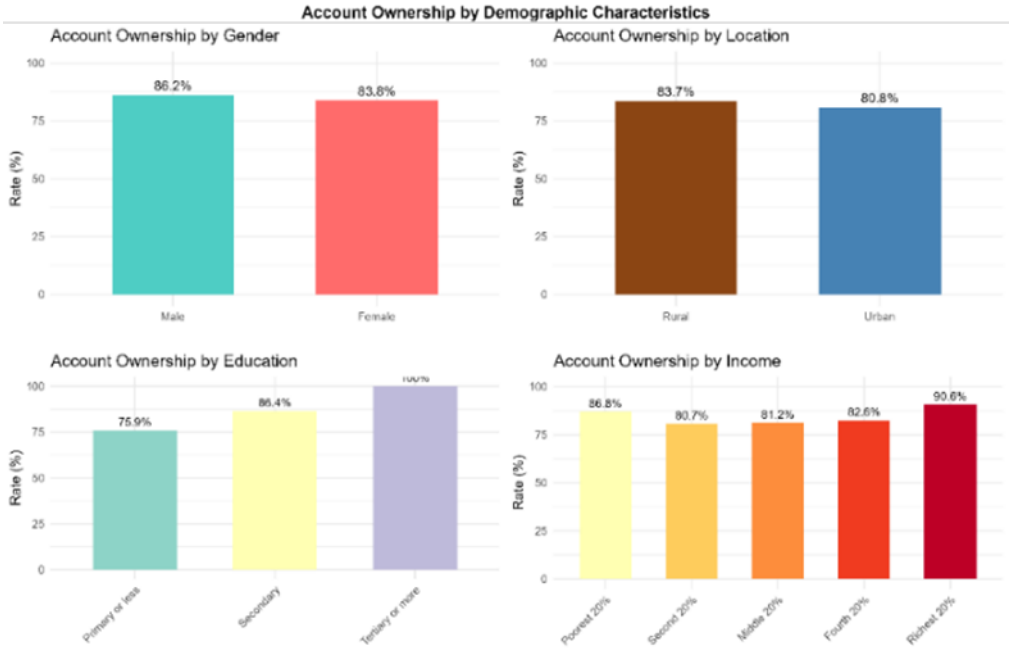


Figure 31: Account Ownership

- Gender & Geographic Parity

The data also shows that there is a very small gap in ownership between the various demographics that typically have a large disparity in ownership between regions—males have account ownership at 86.2% and females come in closely behind at 83.8%. Additionally, when observing the geographic regions within the country, there are no barriers for miles if anything, there is a higher rate of ownership with rural regions at 83.7% compared to urban regions which are at 80.8%. This shows that the distribution of banking infrastructure is available and attainable regardless of gender or physical location.

- Education as a Primary Driver

The level of financial education or degree of education completed is the most significant predictor of whether someone has access to their finances via an account. There is a growing increase in ownership as degree of completed educational level climbs, from 75.9% for Primary or less to 86.4% for Secondary. Moreover, having completed Tertiary or more, we find individuals with a 100% average ownership of an account. This represents a direct correlation between having completed higher learning and a formal financial establishment.

- Income Distribution and Resilience:

The richest income category (Richest 20%) has 90.6% of its members owning assets, however, notable passage has occurred in the lower-income brackets. When reviewing the

asset ownership between traditional and non-traditional asset categories, we find that the poorest income category (Poorest 20%) owns 86.8% of their assets, making them an anomaly in comparison to the three middle-income categories, each with ownership rates between 80.7% and 82.6%. This may indicate that some financial inclusion initiatives are doing a better job than others at identifying and reaching the poorest and most vulnerable consumers in an effective manner, thereby ensuring they do not fall through the cracks when trying to access the formal banking system and build their economic status.

Account Ownership across Age

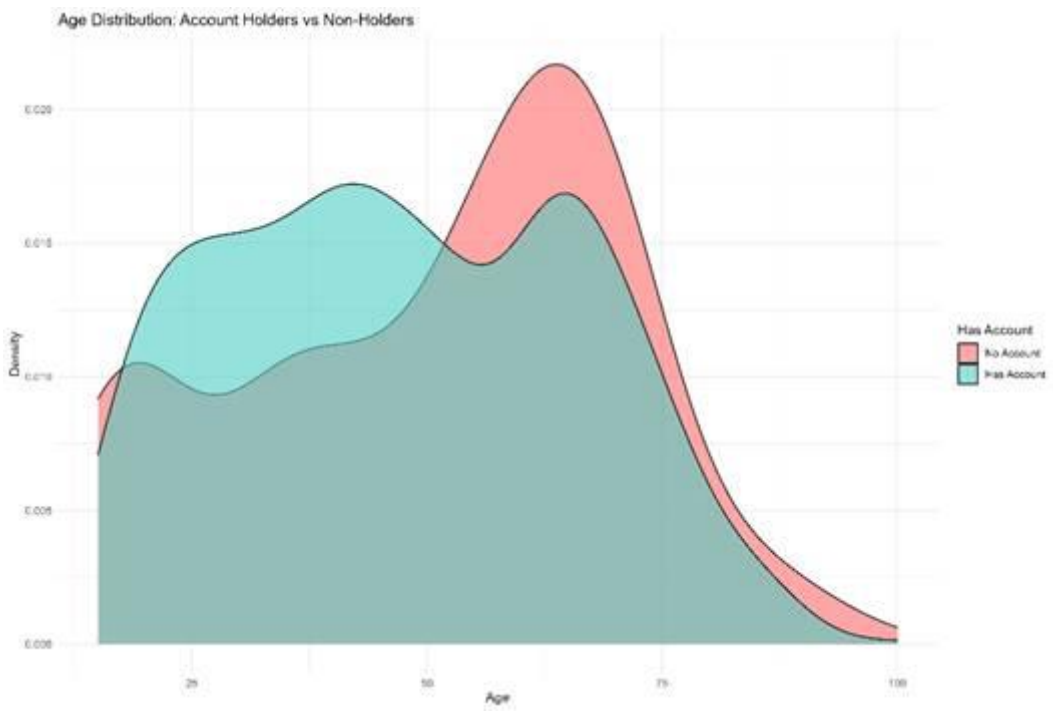


Figure 32:Account Ownership across Age

Between the ages of 25 to 55 is can see the highest number of individuals with accounts relative to those who do not. The "Has Account" density inside that age group is greater than the density of account holders represented by the "No Account". This suggests that employment, having a formal source of income and the management of family finances are all creating a need for banking services among this age group.

As individuals near retirement age (55), there is a sharp increase in the density of accounts held within this group, thereby suggesting a strong need for banking services during working years. However, once an individual reaches the age of 55, the density of individuals without an account begins to increase and eventually surpasses the density of individuals with an account. Individuals without an account reach their highest density of 60 to 70. This indicates that the elderly as a group have been excluded from the formal financial system.

The two distributions begin to decline beginning at the age of 75. This is due to the general population beginning to decline. However, the gap between those with accounts and without can still be seen, with non-account holders representing a larger percentage of the senior

population than those with accounts. This indicates that once seniors have exited the financial system in their sixties, they are less likely to return to the financial system later in life and demonstrate that the elderly are the most financially vulnerable group in this dataset.

Digital Financial Inclusion

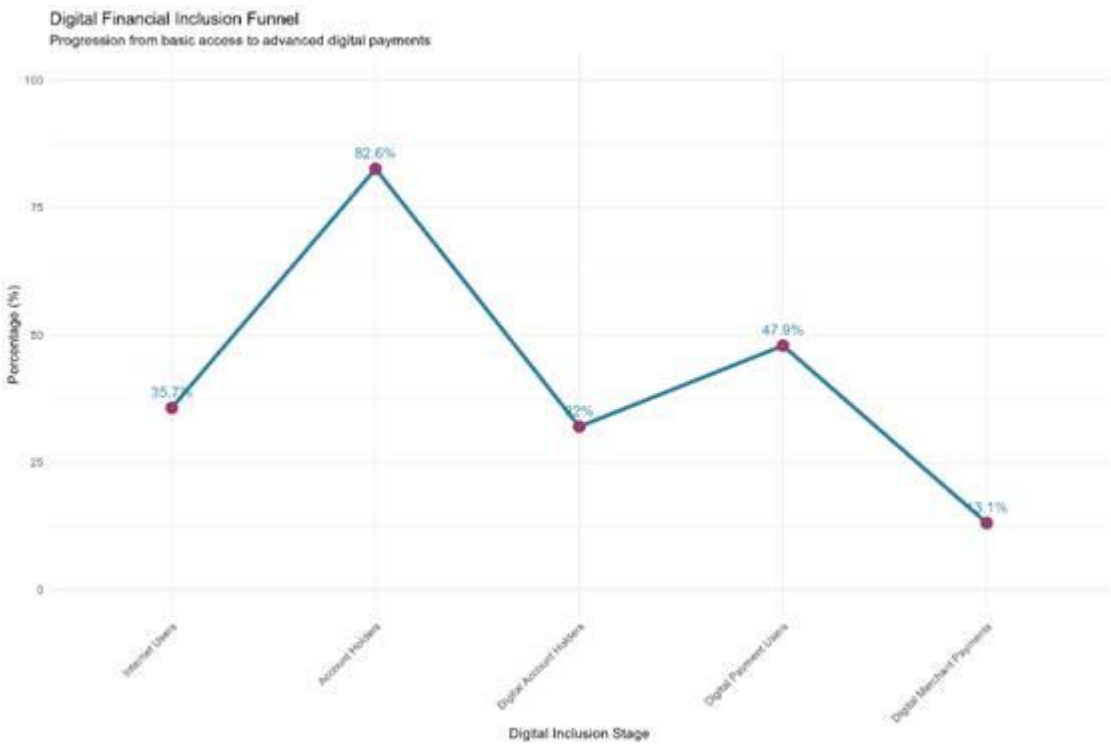


Figure 33:Digital Financial Inclusion

The funnel structure breaks down into three main sections.

Of the general adult population, only 35.7% have internet access. However, 82.6% of the same general population are account holders. Thus, most of the adult population continues to hold traditional, offline accounts.

The second section of the funnel features the Digital Friction Point, which represents a significant drop in penetration from the account holder stage to the digital account holders' stage (32%). We conclude that Few of account holders cannot transition to a digital format from their current, traditional accounts because they either do not have access to the internet or are not sufficiently literate regarding using the internet to perform digital transactions.

The third section of the funnel contains a small area of recovery from stage two (Digital Payment Users = 47.9%). What appears to indicate is that there are some individuals that utilize very limited use of digital payments but do not own any personal digital account.

The Final Advanced Usage Drop-off: The last major decline is in “Digital Payments from Merchants” where 13% of Payments are digital but merchants are using digital payment methods in stores less frequently than other means of payments.

Digital Financial across Location and Education

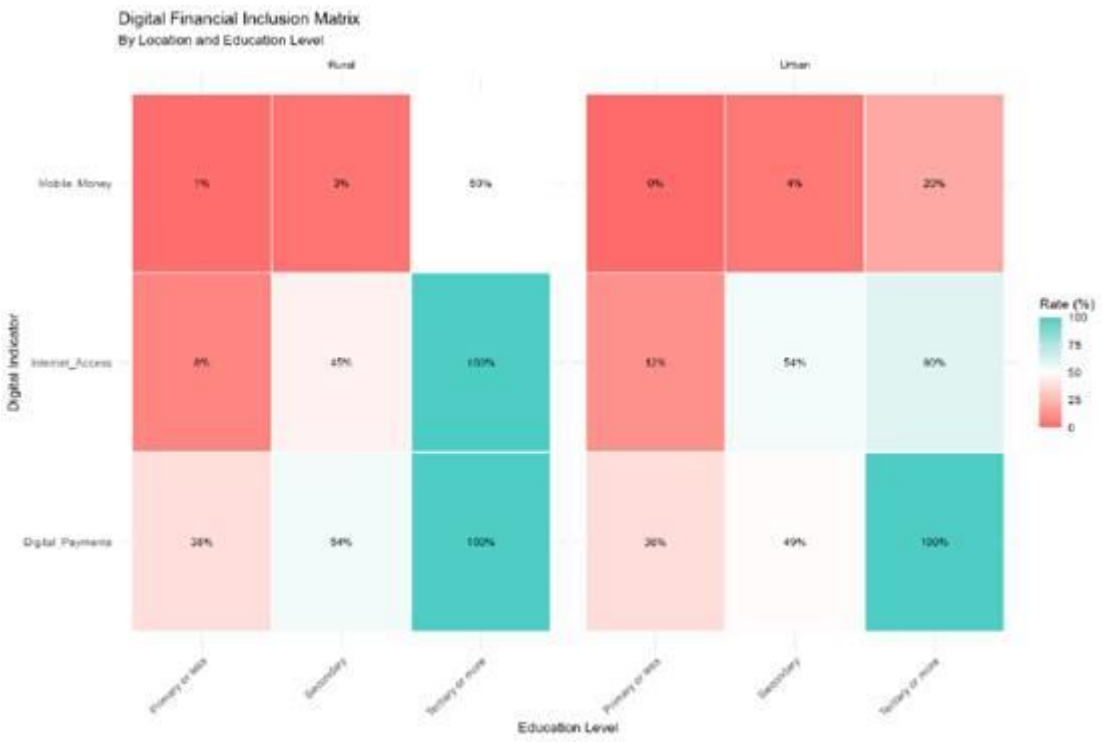


Figure 34:Digital Financial across Location and Education

Inclusion is highly predicted by education because individuals with tertiary education in both rural and urban settings show the highest digital tool usage rates.

Mobile money usage among individuals with primary or secondary education is extremely low (0-4%), but there is a huge increase in mobile money usage for rural individuals with a specific tertiary level of education (50% rural, 20% urban).

The two Indicators of internet access and payment methods are strongly correlated: 100% of rural residents (with tertiary education) have internet connections and use digital payment methods. However, only 60% of urban residents have internet connections when matching both variables (100% use digital payments). This suggests that either a data anomaly exists in urban areas or that urban residents may use an external internet connection for making digital payments.

Key financial indicators

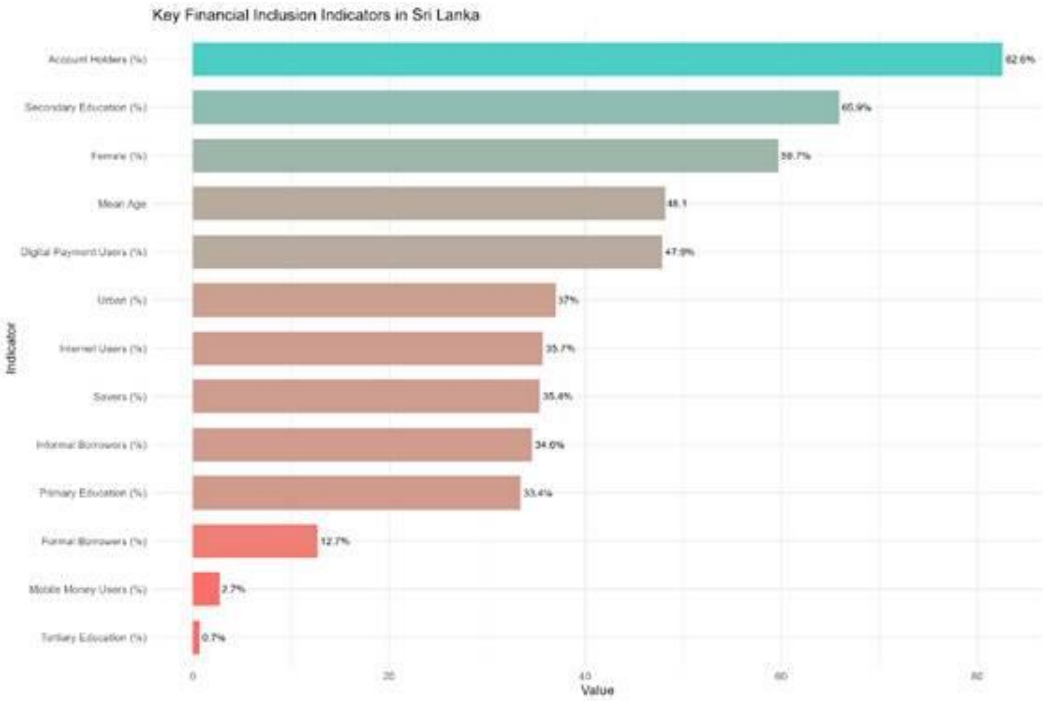


Figure 35:Key financial indicators

Below are the key findings:

- Highs: Access/Reliability & Demographics

Widespread Access: Account Holders at 82.6% suggest that most people in Sri Lanka have basic access to the national formal banking system.

Educational Achievement: 65.9% of the population have achieved High School Education, a solid indicator of financial literacy.

Female/Male Population: The population is very heavily skewed female at 59.7% and relatively old with an average age of 48.1.

- Gaps: Digital/Modern and Usage

Usage Gap: While there are high levels of account ownership, there are significant gaps that exist when measuring the use of digital and modern financial tools.

Digital Divide: Close to ½ of the Sri Lankan population uses digital payments (47.9%) while far fewer (35.7%) use the internet.

Mobile Money: There are very few users of mobile money (2.7%) which implies a significant gap between becoming a bank account owner and becoming a customer service that runs on a mobile device.

6.4.2. Saving Behaviors Across Population Groups

Saving Behavior with Income, Gender and Education

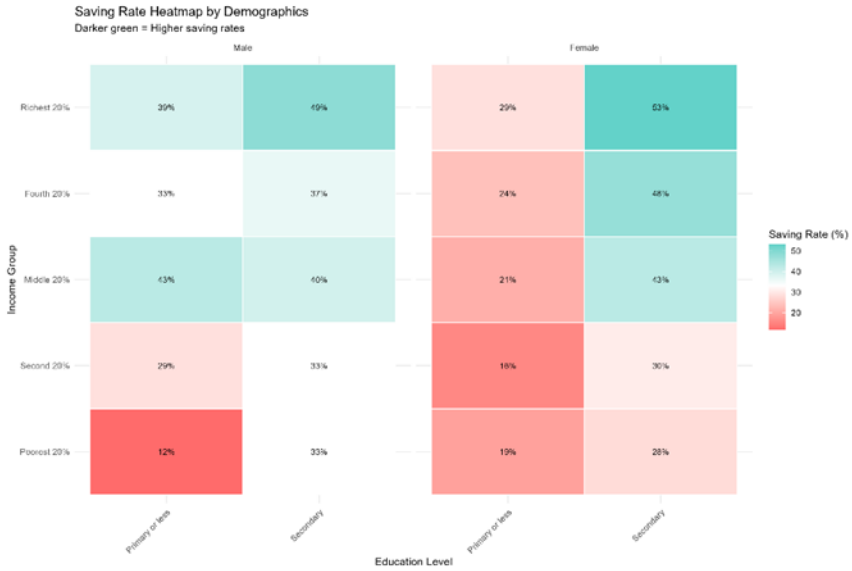


Figure 36: Saving Behavior with Income, Gender and Education

The chart called 'Saving Rate Heatmap by demographics' provides a detailed breakdown of saving behaviors, categorized by gender, income level, and educational attainment. The heat maps, viewed together, indicate that the darker shade of green indicates the highest saving rates, while the lower end of the scale (indicated by shades of red) indicates low saving rates.

- Influence of Education and Gender

Education serves as a primary driver for saving capacity across both genders. For both males and females, individuals with a Secondary education consistently demonstrate higher saving rates compared to those with Primary or less education in almost every income bracket.

- Differences in Savings by Income Level

As would be expected, the data show that as household income increases, savings Rates generally also increase. The Richest 20% of the households have the highest savings Rates (53% for women and 49% for men) while the Poorest 20% of households save at the lowest rates of only 12% (men in the Primary or lower level of education category only). The surprising data is that Middle 20% (men with a Primary education) save at a much higher rate (43%) compared to the richer households in the Fourth 20% of households with Primary education (33%).

Saving Behavior with age and gender

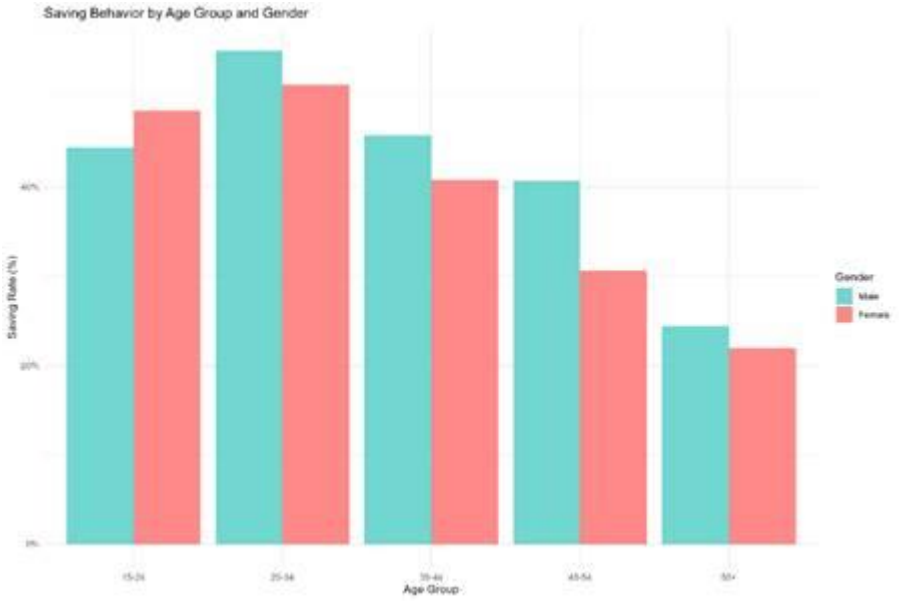


Figure 37: Saving Behavior with age and gender

According to the data, wealth accumulation occurs fastest between the ages of 25 and 34. All respondents from this age group are at their highest rate of saving, with the male average savings rate around 60%, and the female average savings rate very close behind. This possible peak may be because people at this age have established stable careers but may not yet be burdened with the full weight of larger mortgages or the expenses associated with raising children. Even those in the 15 to 24 age range show a relatively high savings rate, most likely because they either live with family or have very low monthly living costs, allowing them to save almost half of their income.

There is a continuing downward trend for both men and women in the 35 to 44 age range and the 45 to 54 age range, with a dramatic drop in both genders' savings rates in the middle years, compared to the savings rates in the earlier years. This downward slope is generally seen as the "middle-age squeeze" where increased household expenditures (such as childcare, education expenses, and housing payments) prevent individuals from saving. By the time an individual is in the 45 to 54 age range, the savings rate of females has dropped below 35%, indicating that the middle years of a person's career are generally the most difficult years to maintain a high degree of liquidity.

When viewed collectively, the 55+ age category displays the least amount of savings in the entire graphic. Although this would seem troubling at first glance, it is to be expected since savers transition from the "accumulation phase" to the "distribution phase" throughout their life cycle.

Saving Behavior Analysis with income

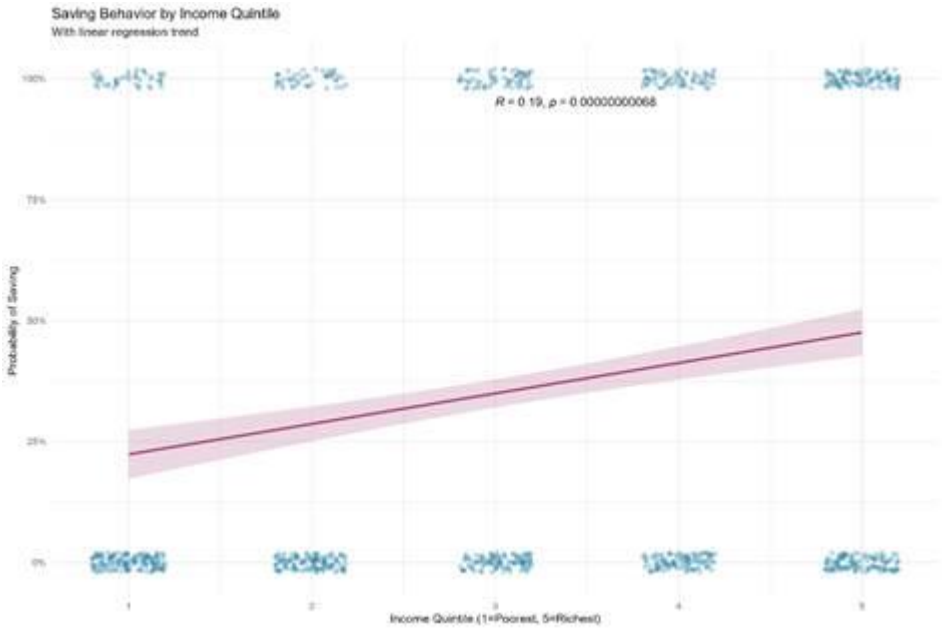


Figure 38:Saving Behavior Analysis with income

The scatter plot titled "Saving Behavior by Income Quintile" shows how a person's income level is related to their likelihood of saving money.

- Correlation Between Income and Saving Money

The most obvious thing we can take away from this scatter plot is that there is a positive linear regression trend as shown by the upwards sloping purple line. As we move from the lowest income quintile (1) to the highest quintile (5), the probability of saving goes up. For example, the lowest income quintile has a probability of saving that is below 25%. However, the highest income has a probability of saving that approaches 50%.

- Statistical Significance vs. Strength

The scatter plot also provides two important characteristic statistical values. They are the R value of 0.19 and the p-value of 0.00000000068.

P-Value: The very low p-value gives us confidence that this relationship is statistically significant, and it is highly unlikely that this upward trend occurred due to chance.

R-Value: An R value of 0.19 indicates a weak correlation between income and saving. Therefore, while income is a contributing factor to whether someone saves money, it is certainly not the only factor. There are many other variables that play a significant role in saving behavior across all income levels, including: cost of living, individual habits, unexpected costs, etc.

Saving Habits of Different Account Types

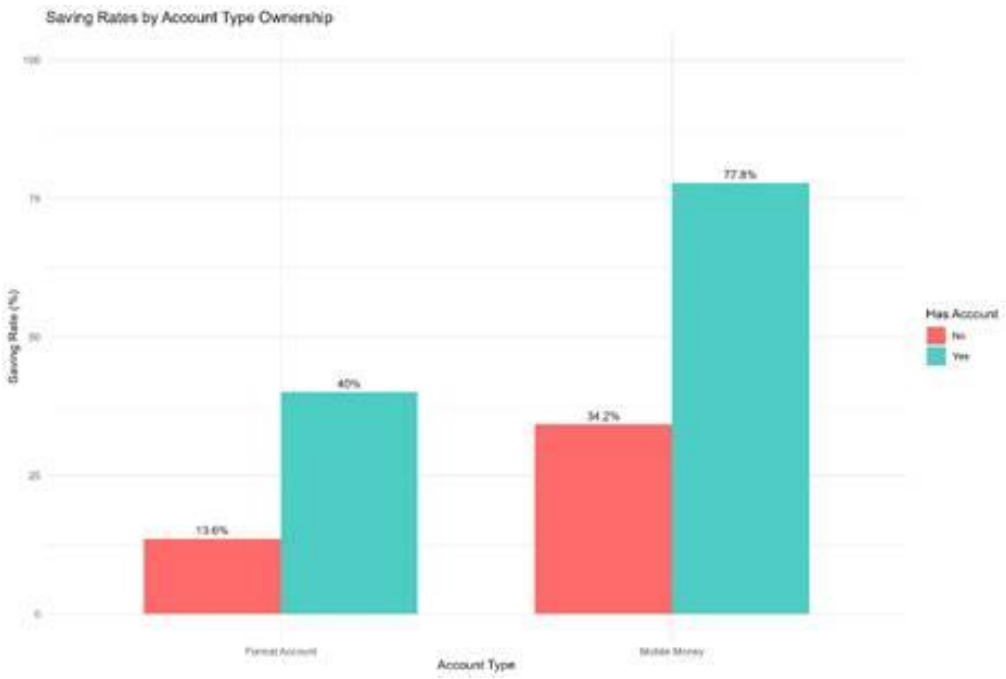


Figure 39:Saving Habits of Different Account Types

The first graph indicates that differences in saving habits can vary significantly based on the type of account held by an individual. When comparing people who do not have any type of account with those who do, there is a dramatic difference. For example, average savings rates for those without a bank account are relatively low (13.6% for bank accounts and 34.2% for mobile accounts); however, due to having an account, there is significant increase in both (formal bank account holders have an average of 40% in savings versus mobile money account holders having an average of 77.8% in savings). This means that Mobile Money is a greater incentive to save than traditional banking – as evidenced by the substantial difference between the savings rates for individuals with a mobile money account and the savings rates of individuals with formal bank accounts. The community, ease of use, and low barriers to entry associated with using digital wallets to save money are much better at building financial reserves than the traditional and often more restrictive characteristics associated with using formal banking institutions.

6.4.3. Role of Mobile Money Accounts in Financial Inclusion

Mobile Money Account Analysis

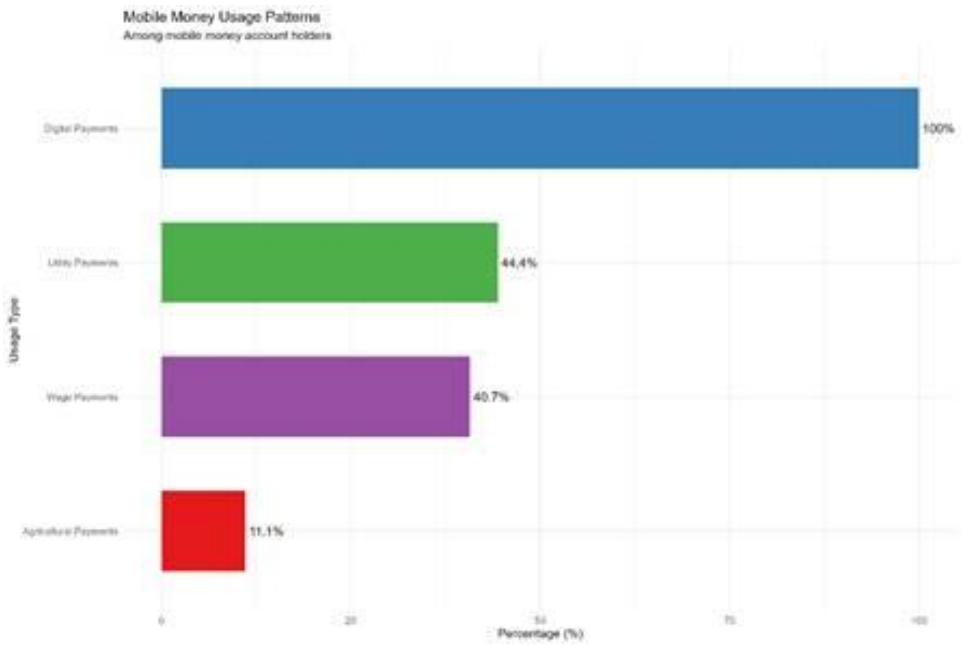


Figure 40:Mobile Money Account Analysis

Use of mobile money has become firmly integrated into the daily routine of mobile money users. The first figure shows that every mobile money user uses mobile money for digital payments – 100% of the users have stated that they perform at least one digital payment using their mobile money service. This indicates that mobile money is now the de facto standard for transferring funds between individuals. Mobile money is not just used for transacting; it is a key platform for managing households and work, as 44.4% of mobile money users have paid their utility bills using their mobile money service, and 40.7% of mobile money users have received their wages through their service. On the other hand, the mobile money data shows that only 11.1% of mobile money users are either making or receiving payment through mobile money in the agricultural industry. This shows that although mobile money is a powerful force in urban and service economies, there is still an opportunity to digitize value chains in the rural and agricultural sectors.

7. Customer Segmentation Analysis (Clustering)

7.1 Overview

A powerful analytical method that groups people according to similar characteristics and behaviors is customer segmentation. In this study, segmentation helps identify unique financial behavior patterns among Sri Lankan customers, enabling targeted product development, marketing strategies, and financial inclusion initiatives.

The findings of a clustering analysis on the Sri Lankan Global Findex 2024 microdata are shown in this chapter. Seven different consumer segments that represent Sri Lanka's varied financial landscape were found by using 65 behavioral variables that were condensed into 21 interpretable components.

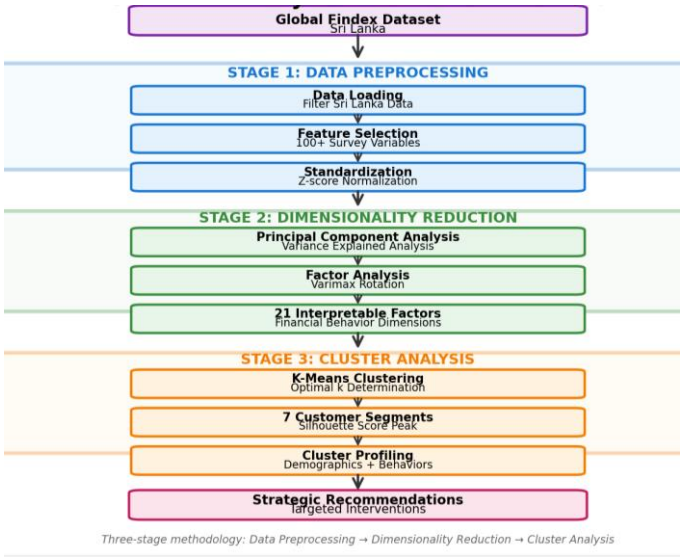


Figure 41:Flowchart for Clustering

This flowchart provides a visual roadmap of the analytical process, showing how raw survey data is transformed through preprocessing, reduced to key behavioral dimensions, and finally segmented into actionable customer groups.

7.2 Methodology

7.2.1 Feature Selection Strategy

Excluded Variables:

The following metadata and demographic variables were excluded from dimensionality reduction to focus on behavioral factors:

- Administrative: year, economy, economycode, regionwb, popadult, wpidrandom, wgt
- Demographics: female, age, educ, incq, empin, urbanicity

These variables serve as profiling characteristics rather than behavioral dimensions for segmentation.

Selected Features:

- Total Features for Analysis: 65 behavioral and financial usage variables
- Variable Types: Binary indicators, categorical responses, and continuous measures

7.2.2 Data Standardization

- Ensure all variables are on comparable scales
- Prevent variables with larger ranges from dominating the analysis
- Required for distance-based algorithms (PCA, K-Means)

7.2.3 Dimensionality Reduction

Before clustering, we reduced the 65 behavioral variables using two complementary techniques:

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was conducted to reduce the dimensionality of the dataset while preserving as much variance as possible.

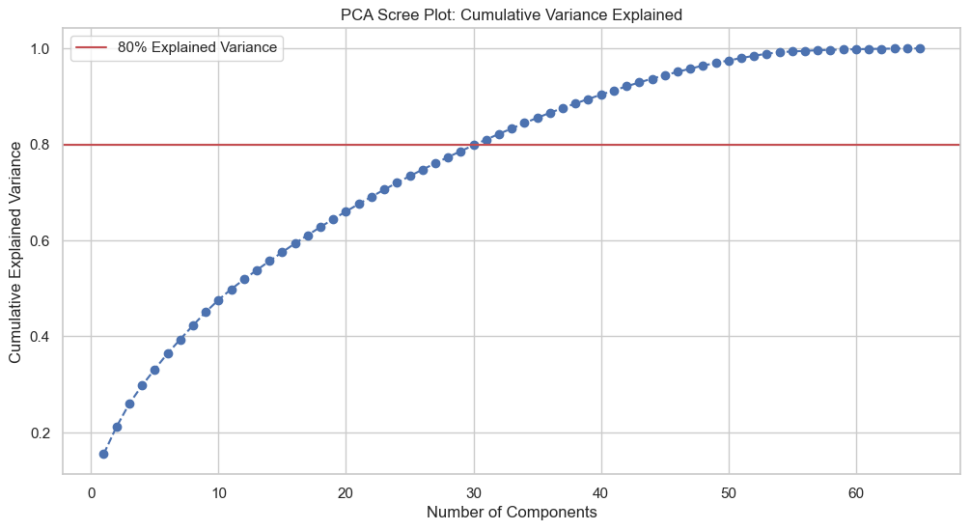


Figure 42:PCA Scree Plot

The gradual accumulation of variance indicates that financial behavior in Sri Lanka is multifaceted, with no single dominant pattern. The selection of 21 components balances parsimony (reducing from 65 variables) with information retention (80% variance explained). This suggests that multiple independent factors drive financial service usage patterns.

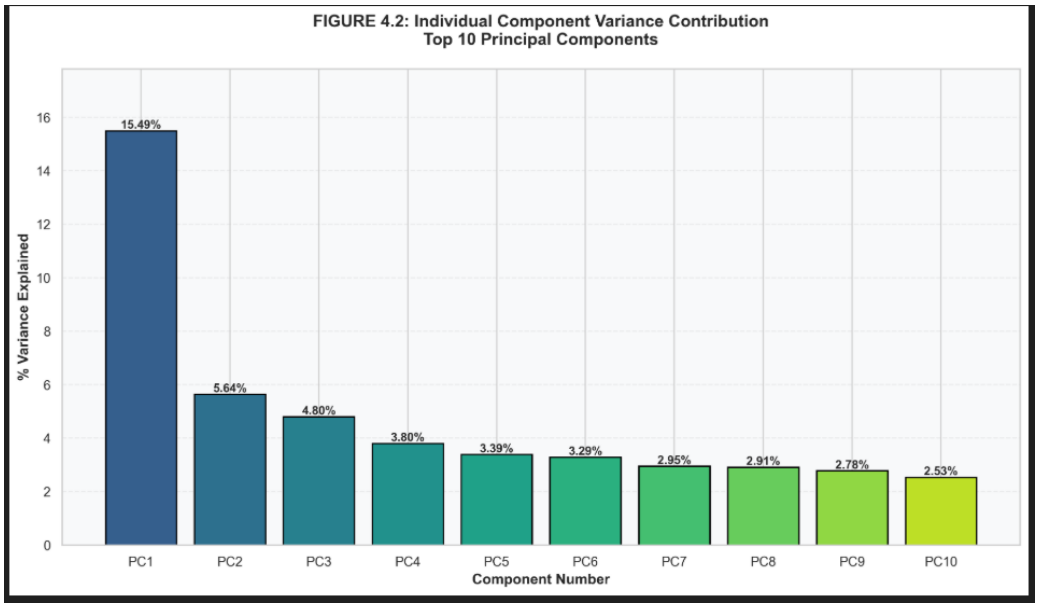


Figure 43:Individual Component Variance Contribution

The first principal component (PC1) captures the largest share of variance at 15.49%, but the remaining components each explain relatively small and gradually decreasing proportions. This indicates that the dataset's structure is complex and multidimensional, requiring multiple components to capture its underlying patterns. The gradual decline in variance explained also suggests that no single component dominates, meaning the original variables are not highly correlated.

Factor Analysis

- Number of Factors: 21 (based on PCA variance analysis)
- Rotation Method: Varimax (orthogonal rotation for interpretability)

Factor Interpretations

Based on top loadings, the 21 factors represent:

Factor 1 - Digital Account Ownership: fin3 (0.89), dig_account (0.89), fin2 (0.63)
Factor 2 - Digital Payments: fin25e2 (0.93), merchantpay_dig (0.83), fin25e1 (0.69)
Factor 3 - Financial Confidence: con14 (0.64), con23 (0.62), con9 (0.59)
Factor 4 - Remittances: domestic_remittances (0.98), fh2 (0.72), fh1 (0.57)
Factor 5 - Traditional Banking: account_fin (0.95), account (0.95), anydigpayment (0.27)
Factor 6 - Transfer Receipts: receive_transfers (0.99), fin37 (0.93), anydigpayment (0.34)
Factor 7 - Wage Income: receive_wages (0.94), fin32 (0.93), anydigpayment (0.25)
Factor 8 - Pension Income: receive_pensions (0.98), fin38 (0.94), anydigpayment (0.22)
Factor 9 - Borrowing Behavior: borrowed (0.90), fin22f (0.71), fin22d (0.44)
Factor 10 - Agricultural Income: receiveagriculture (0.99), fin42 (0.97), accountmob (0.09)
Factor 11 - Utility Payments: pay_utilities (0.98), fin30 (0.93), anydigpayment (0.24)
Factor 12 - Savings Behavior: saved (0.94), fin17a (0.73), fin17c (0.38)
Factor 13 - Digital Literacy: internet_use (0.73), con24 (0.71), con9 (0.34)
Factor 14 - Mobile Money: fin26a (0.60), fin26b (0.56), account_mob (0.42)
Factor 15 - Financial Planning: fin6 (0.68), fin5 (0.61), fin4 (0.23)
Factor 16 - Financial Stress: fin24 (0.59), con12 (0.29), fin25e4 (0.25)
Factor 17 - Credit Access: fin22a (0.60), fin22e (0.32), borrowed (0.30)

Factor 18 - Savings Methods: fin18 (0.61), saved (0.29), fin9b (0.26)
Factor 19 - Account Management: fin8 (0.36), fin22b (0.34), fin22d (0.24)
Factor 20 - Financial Services Usage: fin20 (0.39), con20 (0.32), account_mob (0.25)
Factor 21 - Financial Resilience: con21 (0.27), fh2a (0.23), borrowed (0.22)

Factor Scores

- Output Shape: (1,000 observations × 21 factors)
- These factor scores serve as inputs for the clustering algorithm
- Each respondent has a score on all 21 behavioral dimensions

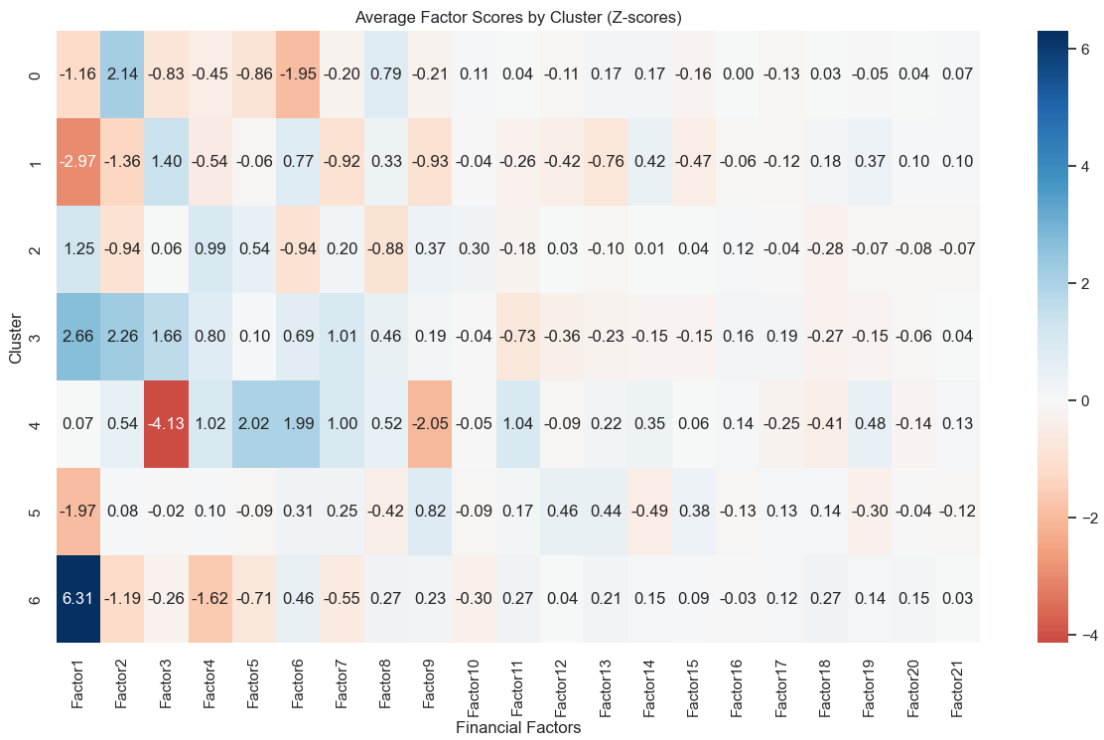


Figure 44:Average Factor Score

This chart shows the results of grouping financial data into 7 different clusters (groups). Each cluster represents companies or data points that share similar financial characteristics. The colors tell us how each group scores on different financial factors - dark blue means very high, dark red means very low, and light colors mean average. For example, Cluster 6 has an extremely high score on Factor 1 (shown by the dark blue color with value of 6.31), while Cluster 1 has a very low score on the same factor (shown in red with value -2.97). This analysis helps identify distinct patterns in financial behavior, making it easier to understand which groups of companies are financially similar and which are different from each other.

7.2.4 Cluster Analysis

K-Means Clustering Algorithm:

- Chosen for its efficiency with large datasets
- Partitions observations into k clusters based on similarity
- Minimizes within-cluster variance while maximizing between-cluster differences

Optimal Cluster Determination

Two complementary methods were employed:

1. Elbow Method
    - Analyzes Within-Cluster Sum of Squares (WCSS/Inertia)
    - Identifies point where marginal improvement diminishes
2. Silhouette Analysis
    - Measures cluster cohesion and separation
    - Range: -1 to +1 (higher is better)
    - Optimal value indicates well-separated clusters

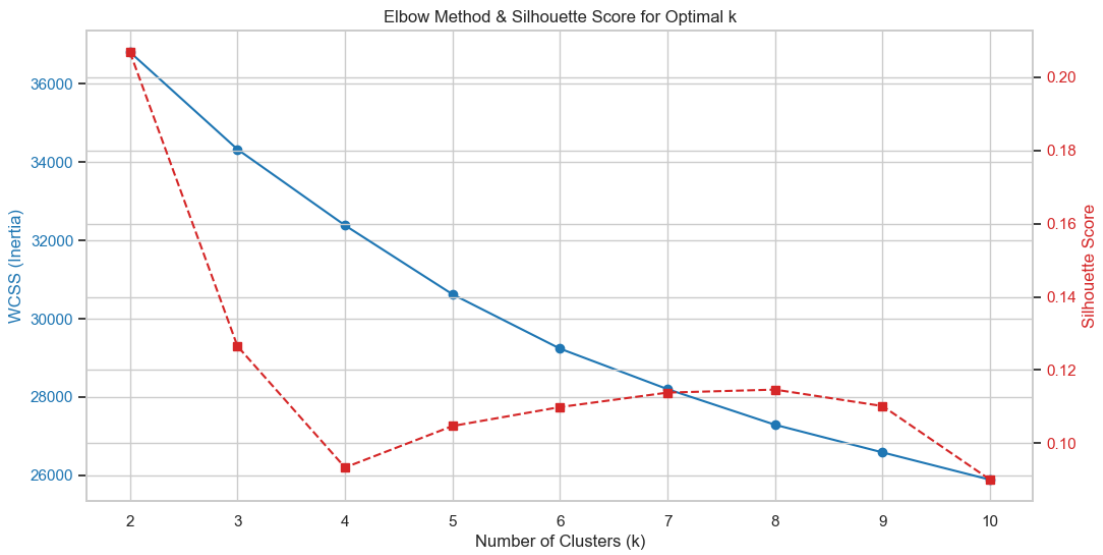


Figure 45:Elbow Method & Silhouette Score for optimal K

This chart helps to find the best number of clusters (groups) for this financial data. It uses two methods:

Blue line (Elbow Method): Shows WCSS (Within-Cluster Sum of Squares) - these measures how tight the groups are. Lower is better, but we want to find where the line starts to flatten out (the "elbow").

Red dashed line (Silhouette Score): Measures how well-separated the clusters are. Higher is better (closer to 1 means better separation).

- The blue line keeps going down, but it bends most sharply around **k=3 to k=4**
- The red line is highest at **k=2** (0.21) and stays relatively stable around **k=7 to k=8** (0.11-0.12)

- After k=7, both lines flatten out

7.3 Descriptive Analysis

7.3.1 Cluster Size Distribution

The 1,000 respondents were distributed across seven clusters as follows:

Cluster	Count	Percentage
Cluster 0	116	11.6%
Cluster 1	167	16.7%
Cluster 2	318	31.8%
Cluster 3	109	10.9%
Cluster 4	134	13.4%
Cluster 5	103	10.3%
Cluster 6	53	5.3%
Total	1,000	100.0%

Table 18:Cluster Size Table

This chart displays how 1,000 people in Sri Lanka are divided into 7 different financial behavior groups. The analysis of the cluster size distribution reveals a population dominated by Non-Agricultural Workers (Cluster 5), which is the largest segment comprising 249 individuals (24.9% of the total).

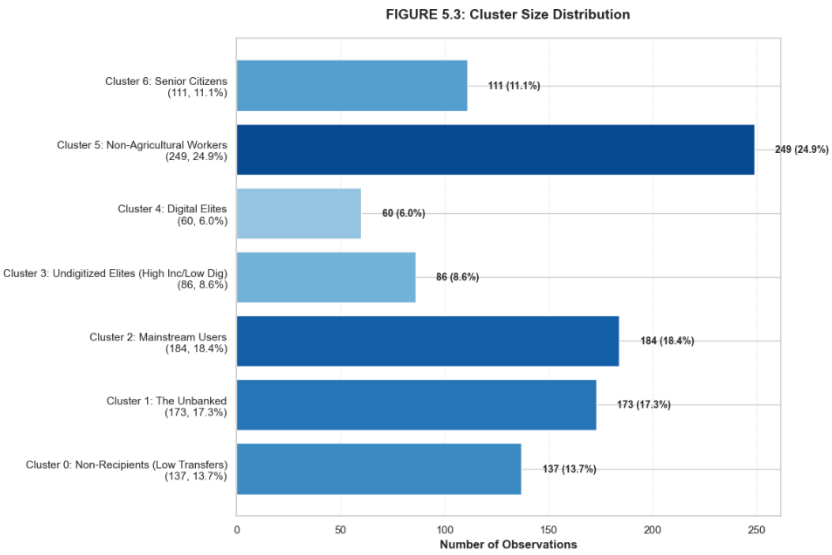


Figure 46:Cluster Size Distribution

The next most significant groups are the Mainstream Users (Cluster 2) excluding 184 individuals (18.4%) and The Unbanked (Cluster 1) with 173 individuals (17.3%), indicating a substantial portion of the population consists of typical financial system users alongside a notable group excluded from traditional banking.

Mean Values by Cluster:

Cluster	Age	Education	Income Quintile	Gender (% Female)	Employment	Urbanicity
0	53.6	1.47	2.28	37.1%	1.58	1.22

1	50.7	1.52	2.73	34.7%	1.66	1.43
2	48.7	1.65	2.86	35.2%	1.67	1.37
3	35.6	2.00	4.08	48.6%	1.37	1.58
4	40.5	1.82	3.54	38.8%	1.54	1.40
5	48.6	1.66	2.98	57.3%	1.31	1.14
6	68.6	1.74	4.02	49.1%	1.74	1.47

Table 19:Mean Values by Clusters

Cluster 0: Non-Recipients (Older - 53.6 years)

This group is comprised of middle-aged and older workers around 54 years old, with a higher proportion of men. They have only a primary-level education and live primarily in rural areas. While they are still working, their income falls into the low-to-middle range, indicating they are likely employed in manual labor or agricultural jobs that provide a modest but below-average living.

Cluster 1: The Unbanked (Older - 50.7 years)

This group consists of middle-aged workers around 51 years old, with a predominantly male composition. They have a primary-level education and are currently employed, earning a moderate, middle-range income. Living in a mix of rural and urban areas, their key defining characteristic is their lack of access to banking, suggesting they operate in a cash-based economy and may face challenges in saving, building credit, or securely managing their earnings.

Cluster 2: Mainstream Users (Middle-aged - 48.7 years)

This group is made up of working adults around 49 years old, with a significant majority being male. They have a secondary-level education and earn a solid, middle-class income. Living across a blend of rural and urban areas, they represent the average worker who has moved beyond a cash-only existence and now utilizes basic banking services to manage their finances.

Cluster 3: Undigitized Elites (Youngest - 35.6 years)

This group is notable for being the youngest segment, around 36 years old, and the most affluent, with the highest income and education levels. They are predominantly urban-dwelling professionals with a balanced gender split. Despite their youth, wealth, and tech-savvy profile, they stand out for their reluctance to adopt digital banking, preferring traditional or alternative methods to manage their substantial finances.

Cluster 4: Digital Elites (Younger - 40.5 years)

This group consists of younger professionals around 41 years old, with a higher proportion of men. They are well-educated, enjoy an upper-middle-class income, and are actively working. Living in a blend of rural and urban areas, they distinguish themselves as the "Digital Elites”

tech-savvy individuals who are comfortable embracing digital banking and financial technology to manage their affluent lifestyles.

Cluster 5: Non-Agricultural Workers (Middle-aged - 48.6 years)

This group is defined by working women around 49 years old, with the highest proportion of females across all segments. They have a secondary-level education and earn a solid middle-class income, while also holding the highest employment rate. Living in the most rural areas, these women are likely employed in non-farm jobs, representing a shift in rural economies where women are actively contributing to household income through work outside of traditional agriculture.

Cluster 6: Senior Citizens (Oldest - 68.6 years)

This group consists of older individuals nearing age 69, most of whom are retired. With a balanced gender split and a secondary level of education, they represent a mix of urban and rural communities. They enjoy a comfortable, upper-middle-class income, allowing for a stable and relaxed lifestyle in their retirement years.

7.4 Results

This financial behavior segmentation analysis of Sri Lankan customers, based on the 2024 Global Findex microdata, has successfully identified seven distinct segments that reflect the diverse financial landscape of the country.

By applying dimensionality reduction techniques like PCA and Factor Analysis to 65 behavioral variables, we extracted 21 interpretable financial factors that served as the foundation for K-Means clustering. The optimal solution of seven clusters provides a nuanced understanding of how Sri Lankans interact with financial services.

Cluster	Name	Size	Age	Income	Key Characteristic
0	Non-Recipients	13.7%	53.6	Low-middle	Rural, older workers
1	The Unbanked	17.3%	50.7	Middle	No bank access
2	Mainstream Users	18.4%	48.7	Middle	Basic banking
3	Undigitized Elites	8.6%	35.6	Highest	Rich, no digital
4	Digital Elites	6.0%	40.5	Upper-middle	Tech-savvy
5	Non-Agricultural Workers	24.9%	48.6	Middle	Rural working women
6	Senior Citizens	11.1%	68.6	Upper-middle	Retired, comfortable

Table 20:Key Characteristic of Clusters

7.4.1 Recommendation

Table 21:Recommended Strategies for Clusters

Segment	Recommended Strategy
Digital Elites	Advanced products (investments, credit, wealth management)
Undigitized Elites	Bridge products to digital with concierge service
Mainstream Users	Upskill to digital with incentives and education
Non-Agricultural Workers	Mobile-based savings and credit tailored for rural women
Senior Citizens	Simplified digital interfaces, trust-building, assisted channels
The Unbanked	Entry-level accounts, financial literacy, agent banking
Non-Recipients	Link to formal transfer systems, basic accounts

8.Discussion

This study reveals that Sri Lanka has achieved strong foundational financial inclusion in terms of account ownership and mobile penetration. However, a significant access–usage gap persists. While more than 80% of adults have bank accounts, less than half actively use digital payment services.

The results show that digital payment adoption is primarily driven by institutional channels such as government transfers and wage payments. Everyday consumer usage remains low, indicating behavioral and trust-related barriers rather than infrastructure constraints.

Demographic analysis highlights strong disparities across age, education, and income levels. Younger, highly educated, and high-income individuals are significantly more likely to adopt digital payments. In contrast, elderly and low-income populations remain digitally excluded.

Machine learning results further confirm that internet usage and financial account ownership are the strongest predictors of adoption. This suggests that improving digital literacy and affordable internet access could significantly accelerate financial digitization.

Overall, Sri Lanka stands at a transition stage where digital infrastructure is adequate, but behavioral transformation and targeted financial product design are needed to increase routine digital payment usage.

Overall Conclusion

This study provided a multifaceted analysis of digital payment adoption, borrowing behavior, financial inclusion patterns, and customer segmentation in Sri Lanka using Global Findex microdata.

The findings show that Sri Lanka has achieved high levels of financial access but continues to face challenges in digital usage intensity. Digital adoption is heavily influenced by education, income, internet access, and age. Borrowing behavior follows a life-cycle pattern, while financial inclusion varies significantly across demographic groups.

Customer segmentation analysis identified distinct behavioral clusters, enabling financial institutions to design more personalized financial products.

Policy implications include:

• Expanding digital literacy programs for older populations

• Enhancing affordable internet access in rural areas

• Encouraging merchant-level digital acceptance

• Designing targeted products for low-income groups

By addressing these gaps, Sri Lanka can move toward a more inclusive and digitally integrated financial ecosystem.

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10. Contribution

Student Name	Student Number	Primary Responsibilities
M.C. Madhusanka	PS/2021/078	Report Finalization, Digital Payment Model
K.A.G.D. Marapana	PS/2021/027	Report Finalization, Clustering System, Project Tracking, Graph Alignment
S.P. Wanigasingha	PS/2021/245	Table of Figures, Advanced EDA, Data Preprocessing
H.M.C.J. Bandara	PS/2021/207	Data Preprocessing, Digital Payment Model Report, Graph Alignment
H.M.K.J. Herath	PS/2021/219	General Report, Advanced EDA (Financial Inclusion)
J.A.A.R. Jayaweera	PS/2021/173	Data Preprocessing, EDA Report (Borrowing Behaviors)
R.R.A. Indranath	PS/2021/164	Data Cleaning & Analysis, Table Alignment
J.A.D.D.S. Jayasooriya	PS/2021/005	Data Preprocessing/Cleaning/Analysis, Digital Payment EDA Report
W.J.E.S.V. Weerasinghe	PS/2021/114	Advanced EDA (Borrowing Behaviors), Hyperparameter Tuning
K.G.R. Kaveesha	PS/2021/103	Data Preprocessing, Cleaning, and Analysis, Table Alignment